Arithmetic Transport and Dimensional Obstructions in UNNS Occurrence Geometry

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Abstract

For a reduced rational $x = a/b$ let $D(x)$ be the set of integers $d \geq 2$ with $d^2 \mid a$ and $d \equiv 1 \pmod{b}$. In exponent coordinates over the primes whose squares divide a, $D(x)$ is the set of nonzero points of a box lying in a lattice Λ_b defined by one congruence. The Graver moves of Λ_b span parallelotopes, and their union U is the realised geometry of the state. When the denominator changes the lattice changes, and we ask what survives; we call this question *transport*.

We separate three notions of survival (vertices, cells of the abstract cell complex, and containment of the realised union) and prove exact criteria for each. In the plane the realised union is monotone under every inclusion of relation lattices, for every box (Theorem 7.5). In dimension three this fails: in infinite families closed under diagonal dilation, including targets with no free coordinate, and for a target that is split in no coordinate, the union of the source is not contained in the union of the target (Theorem 8.2 and Proposition 8.5). The failure is carried by cells that touch all coordinate hyperplanes, which we call hard cells. When the target splits along one coordinate, an exact slab criterion decides containment; it replaces the closed-height criterion of an earlier release (Theorem 9.3). Statements proved by finite exact computation are kept separate, with their scopes, from the theorems.

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1 Introduction

Each reduced rational $x = a/b$ has an occurrence set $D(x) = \{d \geq 2 : d^2 \mid a,\ d \equiv 1 \pmod{b}\}$. Writing the divisors in exponent coordinates over the primes whose squares divide a identifies $D(x)$ with the nonzero points of a box intersected with a lattice Λ_b of exponent vectors defined by one congruence (Proposition 2.1). This finite set, with its Graver moves, parallelotopes and their union, is the geometry studied in the GEO-I releases of the UNNS programme [4]: its Graver moves, the parallelotopes they span, and the union of those parallelotopes.

The question of this manuscript is what happens when the arithmetic moves. Under the collective map $x \mapsto \mathcal{C}(x) = x + L^2/G$, with $L = \operatorname{lcm} D(x)$ and $G = \gcd\{d - 1 : d \in D(x)\}$ (the rational dynamics FIELD-I of the programme [2]), the occurrence set can only grow, $D(x) \subseteq D(\mathcal{C}(x))$, but the lattice and its Graver set change, and geometric structure that existed before need not exist afterwards, nor conversely. We call the study of such comparisons *transport*. The manuscript treats it for explicitly defined source and target lattices and a fixed box; the evolution of whole orbits is the subject of a later manuscript.

Three objects. For a lattice Λ and box Q we use three objects that are *not* interchangeable: the abstract complex $K_{\mathrm{abs}}(\Lambda, Q)$ of cells, the realised union $U(\Lambda, Q) \subset \mathbb{R}^k$ of the corresponding closed parallelotopes, and, when the cells meet only in common faces, the embedded subdivision K_{emb}. Every statement below names the object it concerns.

Results.

(I) *Exact criteria (Sections 3–4).* Inclusion of relation lattices is decided from finitely many group data (Theorem 3.2). A cell of the source survives in the abstract target complex exactly when its vertices lie in the target fibre and its generators are Graver moves of the target (Theorem 4.3); hence no cell of dimension larger than the rank of the common Graver moves survives. For the three-stage orbit $P^4/3 \to 2P^4/3 \to P^4$ all positive-dimensional cells are lost for every number $k \geq 2$ of primes (Corollary 4.6), although the occurrence set strictly grows.

(II) *Abstract loss is not geometric loss (Section 6).* Every abstractly lost edge of that orbit still lies in the target union (Theorem 6.1), while some two-dimensional cells leave it (Proposition 6.2); four plausible monotonicity laws are refuted by exact witnesses.

(III) *Rectangular targets and the corner cube (Section 5).* For a target lattice $\bigoplus o_i\mathbb{Z}$ the uncovered region is exactly an open corner cube O; edges never meet it in any dimension, so the union is preserved for $k = 2$ (Theorem 5.2), while for $k = 3$ the failure is governed by two-dimensional cells and, for the even-sum source, occurs exactly when the scaled box has a side ≥ 2 (Theorem 5.4).

(IV) *Planar monotonicity (Section 7).* For every inclusion $\Lambda \subseteq \Lambda'$ of full-rank lattices of $\mathbb{Z}^2$ and every box, $U(\Lambda, Q) \subseteq U(\Lambda', Q)$ (Theorem 7.5). The proof isolates *hard* cells, those that touch every coordinate hyperplane, and shows that in the plane their geometry forces coverage.

(V) *Dimensional obstruction (Section 8).* Containment fails in dimension three, in infinite families closed under diagonal dilation, with a target having no free coordinate (Theorem 8.2); and also for a target, the body-centred lattice, that is not split in any coordinate (Proposition 8.5, finite-exact).

(VI) *Slab criterion (Section 9).* For a target that splits along one coordinate with period k, the union is an explicit stack of planar unions, and containment of a cell is a planar condition. The criterion as first published (closed height) is false; we prove and use the corrected form (Theorem 9.3).

What is not claimed. Statements checked only by finite exact computation are labelled FINITE-EXACT and carry their domain. Nothing is asserted about arbitrary lattices outside the stated families, and no planar theorem is extended to dimension three. Several results are elementary; no priority claim is made. The classification of non-split three-dimensional targets that preserve the union is open (Section 11).

Relation to earlier work. The relation lattices, Graver cells and planar stabilisation theory are inherited from GEO-I [4] and the Foundations volumes [1, 2, 3] and are cited with their hypotheses; the planar stabilisation theorem G12-T8 is used only as a cited prerequisite (Section 7.5). Throughout, `CW-nnnn` denotes the row of the consolidated theorem crosswalk and the second identifier is the source theorem name; appendix A lists all of them.

Reading guide to the figures. The geometric arguments are illustrated where they are made. Figure 1 fixes the three objects; Figures 2 to 4 illustrate the transport of lattices and of abstract cells; Figures 5 and 6 the corner cube; Figure 7 the difference between abstract loss and containment; Figure 8 the planar theorem; Figures 9 to 11 the three-dimensional obstructions; Figures 12 and 13 the slab criterion; Figures 14 and 15 hard cells and the coupled specimen. Each caption states whether the picture is an *exact computation* (coordinates produced and checked in exact rational arithmetic by `scripts/figs_*.py`; the data are in `results/FIGURE_DATA.json`), a *theorem-level construction* (exact coordinates of the construction in a proof) or a *schematic*. Colours: source structures blue, targets teal, surviving relations green, lost relations red, uncovered regions light orange, witnesses magenta stars; every distinction is also carried by line style, hatching or marker shape.

2 Setting and notation

2.1 Fibres of occurrence sets

A *state* is a positive rational $x = a/b$ in lowest terms. Its occurrence set is $D(x) = \{d \geq 2 : d^2 \mid a,\ d \equiv 1 \pmod{b}\}$ (Foundations Vol. I, Thm. 1.2 [1]). Let $\mathcal{P} = \{p_1 < \cdots < p_k\}$ be the primes

with $p_i^2 \mid a$, $q_i = \lfloor v_{p_i}(a)/2 \rfloor \geq 1$, and $Q = \prod_i [0, q_i]$ the *box*. For $\alpha \in \mathbb{Z}^k$ put $\chi_b(\alpha) = \prod_i p_i^{\alpha_i} \bmod b$ (negative exponents through inverses; $\gcd(p_i, b) = 1$ because $\gcd(a, b) = 1$) and

$$\Lambda_b = \ker \chi_b \subseteq \mathbb{Z}^k, \qquad F(\Lambda, Q) = (\Lambda \cap Q) \setminus \{0\}.$$

Proposition 2.1 (fibre)**.** *The map $\alpha \mapsto \prod p_i^{\alpha_i}$ is a bijection $F(\Lambda_b, Q) \to D(x)$. The index is $[\mathbb{Z}^k : \Lambda_b] = |H_b|$ with $H_b = \langle p_1, \ldots, p_k \rangle \leq (\mathbb{Z}/b)^\times$.*

Proof. If $d^2 \mid a$ then every prime factor p of d has $p^2 \mid a$, so $d = \prod p_i^{\alpha_i}$ with $0 \leq \alpha_i \leq q_i$; this exponent vector is unique, and $d \equiv 1 \pmod b$ iff $\chi_b(\alpha) = 1$. The vector $\alpha = 0$ gives $d = 1 \notin D(x)$. The map $\chi_b : \mathbb{Z}^k \to H_b$ is onto with kernel Λ_b. □

For $b = 1$ we have $\Lambda_1 = \mathbb{Z}^k$. A lattice here always means a full-rank sublattice of $\mathbb{Z}^k$ (finite index); this holds for every Λ_b.

2.2 Moves, cells and the three objects

Definition 2.2 (conformal order, Graver set)**.** For $v, w \in \mathbb{Z}^k$ write $w \preceq v$ if $w_i v_i \geq 0$ and $|w_i| \leq |v_i|$ for all i. For a lattice Λ, $\mathrm{Gr}(\Lambda)$ is the set of $\preceq$-minimal nonzero elements of Λ (the *Graver set*); it is symmetric under $v \mapsto -v$ (Graver [7]; see also [8, 9]). For $q \in \mathbb{Z}^k_{>0}$ the *region Graver set* $\mathrm{Gr}_q(\Lambda)$ is the set of $\preceq$-minimal nonzero elements of $\Lambda \cap [-q, q]$.

Definition 2.3 (cell)**.** Let $g_1, \ldots, g_j \in \mathrm{Gr}_q(\Lambda)$ be linearly independent and $u \in \mathbb{Z}^k$. The parallelotope $\Pi(u; g_1, \ldots, g_j) = u + \sum_i [0, 1] g_i$ is a *j-cell of* (Λ, Q) if all its 2^j corners lie in $F(\Lambda, Q)$. (A 0-cell is a point of F.) The *abstract complex* $K_{\mathrm{abs}}(\Lambda, Q)$ is the set of all cells, a cell being identified with its vertex set (Lemma 2.4); it is closed under taking faces. The *realised union* is $U(\Lambda, Q) = \bigcup$ of the closed cells. The *origin-allowed union* $U^0(\Lambda, Q)$ is defined in the same way with the corners allowed to be 0.

The origin is excluded because $\alpha = 0$ corresponds to $d = 1 \notin D(x)$. This single exclusion is the source of every phenomenon in this manuscript. A third object, the *embedded subdivision* K_{emb}, exists only when any two closed cells meet in a common face; it is never needed below and when a statement is about K_{abs}, U or K_{emb} it says so. Boundaries of cells in K_{abs} are $\partial\Pi(u; g_1..g_j) = \sum_i (-1)^{i-1} [\Pi(u + g_i; \widehat{g_i}) - \Pi(u; \widehat{g_i})]$; $\partial\partial = 0$.

Lemma 2.4 (cells are intrinsic)**.** *Order $\mathbb{Z}^k$ lexicographically. A cell c with vertex set $V(c)$ has its lexicographically smallest vertex $m(c)$ as a base point and the set* $\mathrm{Gen}(c)$ *of its generators, taken lexicographically positive, equals the set of edge vectors of* $\operatorname{conv} V(c)$ *at $m(c)$. In particular $V(c)$ determines c and* $\mathrm{Gen}(c)$*.*

Proof. $\operatorname{conv} V(c) = \Pi(u; g_1..g_j)$ is a j-dimensional parallelotope; its vertices are $u + \sum_{i \in S} g_i$. The lexicographic order is translation invariant, so the smallest vertex is $u + \sum_{g_i < 0} g_i$, and the edges at that vertex of the parallelotope are along $\pm g_i$ leaving in the lexicographically positive direction. Edges of $\operatorname{conv} V(c)$ depend only on the convex body, hence only on $V(c)$. □

Remark: since $\mathrm{Gr}_q(\Lambda) = -\mathrm{Gr}_q(\Lambda)$, "$\mathrm{Gen}(c) \subseteq \mathrm{Gr}_q(\Lambda)$" does not depend on orientations.

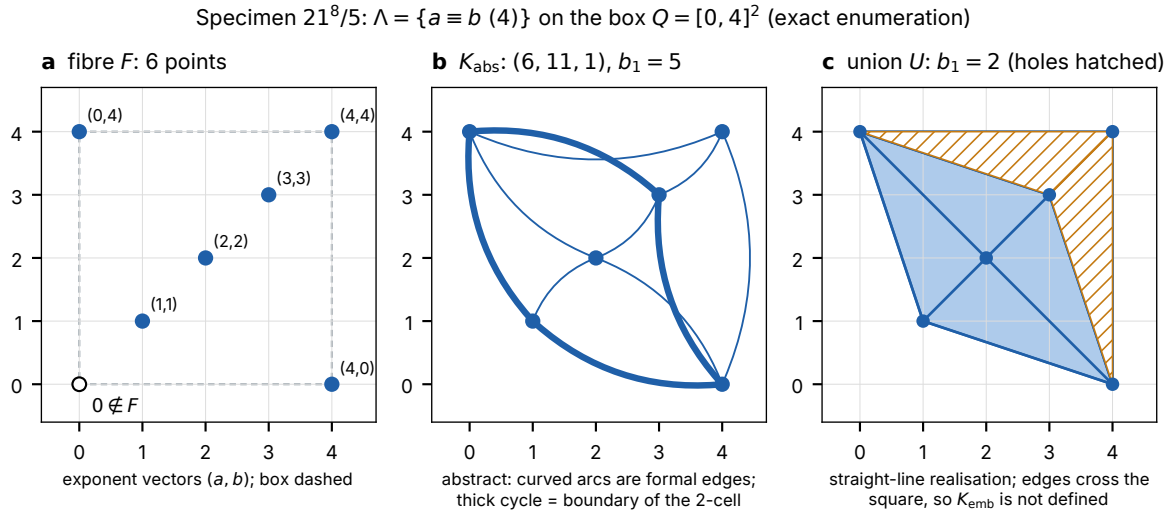


Figure 1: The three objects for the specimen $21^8/5$ ($\Lambda = \{a \equiv b\ (4)\}$, $Q = [0,4]^2$; *exact computation*). (a) The fibre $F(\Lambda, Q)$: six points, the origin being excluded. (b) The abstract complex K_{abs}: f-vector $(6, 11, 1)$, Betti numbers $(1, 5, 0)$; edges are drawn as arcs to stress that they are formal incidences, and the thick cycle bounds the single 2-cell. (c) The realised union U of the closed cells with straight edges: it has two bounded complementary components (hatched), so $b_1(U) = 2 \neq b_1(K_{\mathrm{abs}}) = 5$. Edges of K_{abs} cross inside the square, so closed cells do not all meet in common faces and K_{emb} is *not* defined for this specimen. The Betti numbers agree with those reported by the release; $b_1(U)$ was obtained here with a floating-point polygon routine on exact integer coordinates (Definition 2.3, Proposition 6.3).

2.3 Notation table

symbol	meaning
$x = a/b$; $D(x)$	state; occurrence set
$\mathcal{P} = \{p_i\}$, $Q = \prod[0, q_i]$	active primes; exponent box
Λ_b, H_b, χ_b	relation lattice, group generated by the primes mod b, character
$F(\Lambda, Q)$	fibre $(\Lambda \cap Q) \setminus \{0\}$
$\mathrm{Gr}(\Lambda)$, $\mathrm{Gr}_q(\Lambda)$	Graver set; region Graver set (the source writes G; G is also the gcd of FIELD-I)
$\preceq$	conformal order
$\Pi(u; g_1..g_j)$, K_{abs}, U, U^0	cell (source: $P(u; \dots)$), abstract complex, union, origin-allowed union
E_k	even-sum lattice $\{z \in \mathbb{Z}^k : \sum z_i \text{ even}\}$ (index 2)
$\mathcal{C}$	collective map $x \mapsto x + L^2/G$ with $L = \mathrm{lcm}\, D(x)$, $G = \gcd\{d - 1\}$
v^+, v^-	$\max(v, 0)$, $\min(v, 0)$ coordinatewise

2.4 Model boundaries

The objects here belong to the model classes M7 (lattice and target transport) and, for the arithmetic inputs, M1 (FIELD-I collective map) and M2/M3 (GEO-I occurrence geometry, planar model). The binary 0/1 obstruction model (M4), the rational-state move complexes $K_{\mathrm{axis}}, K_{\mathrm{move}}$ (M5) and the history complexes (M6) are different objects and are not used. The

map $\mathcal{C}$ is one specified collective rule; it is not forced by the arithmetic formula for $D(x)$. A *transition* $x \to x' = \mathcal{C}(x)$ is called *admissible* if x' has the same active primes and box; only admissible transitions are compared here.

3 Transport of relation lattices

Throughout this section $p_1 < \cdots < p_k$ are primes coprime to the moduli b, b' and $\Lambda_b = \ker \chi_b$.

3.1 Lattice order from group orders

Proposition 3.1. $[\mathbb{Z}^k : \Lambda_b] = |H_b|$; $\Lambda_b \cap \Lambda_{b'} = \Lambda_{\mathrm{lcm}(b,b')}$; *hence* $\Lambda_b \subseteq \Lambda_{b'} \iff |H_{\mathrm{lcm}(b,b')}| = |H_b|$.

Source record: `E2-2` (`CW-0457`); status PROVED; treatment FULL.

Proof. $\chi_b : \mathbb{Z}^k \to H_b$ is onto with kernel Λ_b, giving the index. By the Chinese remainder theorem $\chi_b(\alpha) = 1 = \chi_{b'}(\alpha)$ iff $\chi_{\mathrm{lcm}}(\alpha) = 1$. If $\Lambda_b \subseteq \Lambda_{b'}$ then $\Lambda_{\mathrm{lcm}} = \Lambda_b$, so the indices agree. Conversely $\Lambda_{\mathrm{lcm}} \subseteq \Lambda_b$ with equal finite index gives equality, so $\Lambda_b = \Lambda_{\mathrm{lcm}} \subseteq \Lambda_{b'}$. □

3.2 A decision criterion

Theorem 3.2 (lattice transport)**.** *Let* $m_i = \mathrm{ord}_b(p_i)$. *The following are equivalent: (a)* $\Lambda_b \subseteq \Lambda_{b'}$; *(b)* $p_i \bmod b \mapsto p_i \bmod b'$ *extends to a well-defined homomorphism* $H_b \to H_{b'}$; *(c) every element of the finite set* $\{m_i e_i\} \cup \{\alpha \in \prod[0, m_i) : \chi_b(\alpha) = 1\}$ *lies in* $\Lambda_{b'}$.

Source record: `P2-T1` (`CW-0467`); status PROVED; treatment FULL.

Proof. A homomorphism φ with $\varphi \circ \chi_b = \chi_{b'}$ exists iff $\ker \chi_b \subseteq \ker \chi_{b'}$ (universal property of the surjection $\chi_b : \mathbb{Z}^k \to H_b$); this is (a) $\iff$ (b). For (c): $m_i e_i \in \Lambda_b$. Any $\alpha \in \mathbb{Z}^k$ is $\sum_i t_i m_i e_i + r$ with $r \in \prod[0, m_i)$, and $\alpha \in \Lambda_b$ iff $r \in \Lambda_b$; so the displayed set generates Λ_b, and $\Lambda_{b'}$ is a group. □

Corollary 3.3 (denominators)**.** *If* $b' \mid b$ *then* $\Lambda_b \subseteq \Lambda_{b'}$. *This is sufficient, not necessary, and does not give strict inclusion.*

Source record: `P2-T1` (`CW-0467`); status PROVED; treatment FULL.

Proof. A congruence modulo b implies the same congruence modulo b'. For non-necessity take $b = 5$, $b' = 4$, primes $3, 7$: $\Lambda_5 = \{a \equiv b\ (4)\}$ and $\Lambda_4 = \{a \equiv b\ (2)\}$ (see the proof of Theorem 4.8), so $\Lambda_5 \subsetneq \Lambda_4$ though $4 \nmid 5$ and $5 \nmid 4$. Equality can occur: for the primes $\{5, 17, 29\}$ one has $5, 17, 29 \equiv -1 \pmod 3$ and $\equiv 5 \pmod{12}$ with $5^2 \equiv 1$, so $\Lambda_3 = \Lambda_{12} = \{\alpha : \sum \alpha_i \text{ even}\}$ although $b = 3 \neq 12 = b'$. □

This correction (ER-16) replaces the earlier phrase "denominator divisibility is neither necessary nor sufficient" of the first release; the sufficiency statement is the form used here.

Lemma 3.4 (order rule)**.** *Let* $n_i = \mathrm{ord}_{b'}(p_i)$. *If* $g \in \mathbb{Z}^k$ *has* $|g_i| \geq n_i$ *for some* i *and* $g \neq \pm n_i e_i$, *then* $g \notin \mathrm{Gr}(\Lambda_{b'})$.

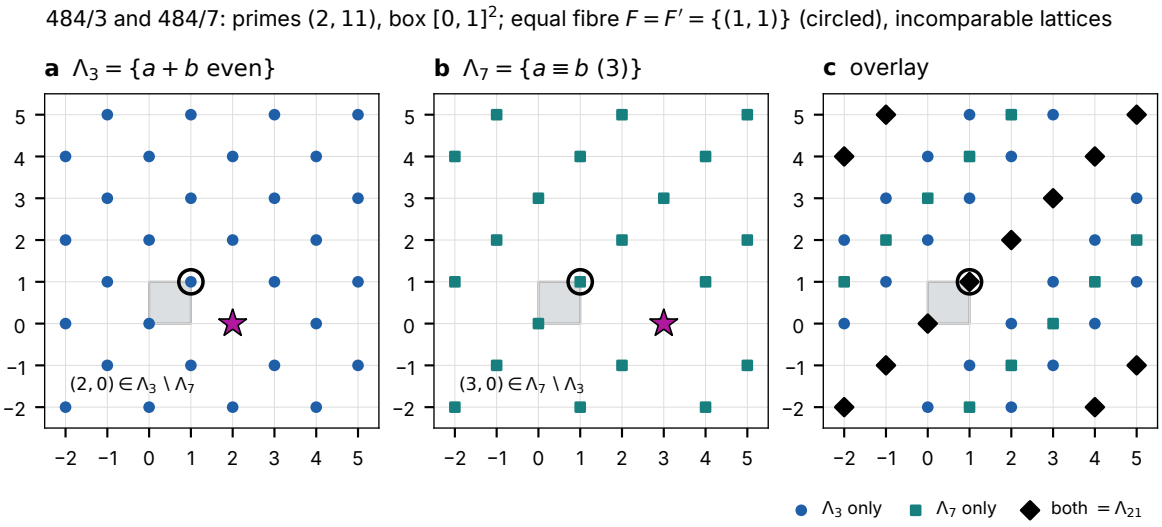


Figure 2: Equal fibres, incomparable lattices (Proposition 3.5(b); *exact computation*): $x = 484/3$ and $x' = 484/7$, primes $(2, 11)$, box $[0, 1]^2$, window $[-2, 5]^2$ of the infinite lattices. (a) $\Lambda_3 = \{a + b \text{ even}\}$; (b) $\Lambda_7 = \{a \equiv b\ (3)\}$; (c) overlay, with $\Lambda_3 \cap \Lambda_7 = \Lambda_{21}$. The shaded unit square is the box and the point $(1, 1)$ is the common fibre $F = F'$. The stars are the exact witnesses $(2, 0) \in \Lambda_3 \setminus \Lambda_7$ and $(3, 0) \in \Lambda_7 \setminus \Lambda_3$: the lattices are incomparable although the fibres coincide, so the spanning hypothesis in Proposition 3.5(c) cannot be dropped.

Source record: `P2-L2` (`CW-0468`); status PROVED; treatment FULL.

Proof. $h = \operatorname{sign}(g_i)\, n_i e_i \in \Lambda_{b'}$ is nonzero, $h \preceq g$ and $h \neq g$. □

This is an $O(k)$ *necessary* condition for a vector to survive as a target Graver move; it is not sufficient.

3.3 Fibre inclusion versus lattice inclusion

Proposition 3.5. *(a) If $\Lambda \subseteq \Lambda'$ and the box is common, then $F(\Lambda, Q) \subseteq F(\Lambda', Q)$. (b) The converse is false: $x = 484/3$ and $x' = 484/7$, primes $(2, 11)$, box $[0, 1]^2$, have $F = F' = \{(1, 1)\}$, yet $\Lambda_3 = \{a + b \text{ even}\}$ and $\Lambda_7 = \{a \equiv b\ (3)\}$ are incomparable. (c) If $F \subseteq F'$ and $\operatorname{span}_{\mathbb{Z}} F = \Lambda$, then $\Lambda \subseteq \Lambda'$.*

Source record: `E2-4` (`CW-0459`); status PROVED; treatment ADAPTED.

Proof. (a) is immediate. (c): $F \subseteq F' \subseteq \Lambda'$; take the span. (b): $484 = 2^2 \cdot 11^2$, so $d^2 \mid 484$ iff $d \mid 22$; $22 \equiv 1$ modulo 3 and 7, while $2, 11 \not\equiv 1$; thus $D = D' = \{22\}$, $F = F' = \{(1, 1)\}$ for the box $[0, 1]^2$. Modulo 3: $2 \equiv 11 \equiv -1$, so $\chi_3(a, b) = (-1)^{a+b}$. Modulo 7: 2 has order 3 and $11 \equiv 2^2$, so $\chi_7(a, b) = 2^{a+2b}$, trivial iff $a \equiv b \pmod 3$. Then $(2, 0) \in \Lambda_3 \setminus \Lambda_7$ and $(3, 0) \in \Lambda_7 \setminus \Lambda_3$. □

The adapted statement (c) is the content of Lemma E2-4 ("no shrinking"): along a collective step with $D(x) \subseteq D(\mathcal{C}(x))$ (Foundations Vol. II, Thm. 1.3, F.T31 [2]; cited, not reproved) one has $F \subseteq F'$, so a spanning fibre forces $\Lambda \subseteq \Lambda'$. Also, every denominator along an orbit divides

$G(x_0)$ (E2-3, resting on the same inherited facts), so the lattices along an orbit all contain $\Lambda_{G(x_0)}$ in fixed coordinates. The spanning hypothesis must accompany every use of (c) (ER-14).

Source record: `E2-3` (`CW-0458`); status PROVED; treatment ADAPTED.

3.4 Corpus statistics (finite-exact)

Proposition 3.6 (FINITE-EXACT; source scope)**.** *In the exact corpus of* 596 *admissible non-EQUAL transitions of the source (*582 *with* $k = 2$*,* 14 *with* $k = 3$*) one finds ENLARGE* 275*, SHRINK* 262*, INCOMPARABLE* 59*; in every row* $F \subseteq F'$*. No SHRINK or INCOMPARABLE row has a spanning source fibre (*321/321*). If* $b \mid b'$ *the class is SHRINK in all* 199 *such rows; if* $b' \mid b$ *it is ENLARGE in all* 154*.*

Source record: `P2-D` (`CW-0472`); status FINITE-EXACT; treatment FINITE.

These numbers are a computation in the stated corpus, reported from the release; the planar sampling windows of the first release were incomplete for boxes with $q_1 < q_2$, and all 326 geometry verdicts were recomputed with full-box certificates in the next release with 0 verdicts changed (ER-18); they are not re-derived here and are not extrapolated. Their consequence for this manuscript is only qualitative: transitions with SHRINK or INCOMPARABLE lattices occur under $\mathcal{C}$ (so "the lattice never shrinks" is not a law, ER-14), and transport theorems must either assume $\Lambda \subseteq \Lambda'$ or avoid lattice comparison.

4 Abstract transport of cells

4.1 Region Graver sets and the survival criterion

Lemma 4.1. *For every lattice* $\Lambda \subseteq \mathbb{Z}^k$*,* $\mathrm{Gr}_q(\Lambda) = \{g \in \mathrm{Gr}(\Lambda) : |g_i| \le q_i \ \forall i\}$*.*

Source record: `P2-L1` (`CW-0466`); status PROVED; treatment FULL.

Proof. If $h \preceq g$ and $|g_i| \le q_i$ then $|h_i| \le q_i$, so every conformal subvector of a vector of the box lies in the box; minimality within the box and within Λ coincide for vectors of the box. □

Theorem 4.2 (generator survival without the target complex)**.** *Let* $g \in \mathrm{Gr}_q(\Lambda)$*. Then* $g \in \mathrm{Gr}_q(\Lambda')$ *iff* $g \in \Lambda'$ *and no* $v \in \Lambda' \setminus \{0, g\}$ *satisfies* $v \preceq g$*. Only* $\prod_i (|g_i| + 1) - 2$ *candidate vectors* v *have to be inspected, and only* $\chi_{b'}$ *is needed.*

Source record: `P2-T2` (`CW-0469`); status PROVED; treatment FULL.

Proof. Definition of conformal minimality and Lemma 4.1; the candidates are the integer points of the conformal box of g. □

Theorem 4.3 (abstract-cell transport)**.** *Let* Λ, Λ' *be lattices,* Q *a common box, and* c *a cell of* $K_{\mathrm{abs}}(\Lambda, Q)$*. Then*

$$c \in K_{\mathrm{abs}}(\Lambda', Q) \iff V(c) \subseteq F(\Lambda', Q) \text{ and } \mathrm{Gen}(c) \subseteq \mathrm{Gr}_q(\Lambda').$$

Put $S = \mathrm{Gr}_q(\Lambda) \cap \mathrm{Gr}_q(\Lambda')$*. Then (1) if* $\Lambda \subseteq \Lambda'$ *all* 0*-cells persist, and a* j*-cell* c *persists iff* $\mathrm{Gen}(c) \subseteq S$*; (2) no cell of dimension* $> \mathrm{rank}\langle S \rangle$ *persists; in particular* $S = \emptyset$ *implies that all cells of positive dimension are lost, and* $\mathrm{rank}\langle S \rangle = 1$ *implies that all* 2*-cells are lost; (3) if* $\mathrm{Gr}_q(\Lambda) \subseteq \mathrm{Gr}_q(\Lambda')$ *and* $\Lambda \subseteq \Lambda'$ *then* $K_{\mathrm{abs}}(\Lambda, Q) \subseteq K_{\mathrm{abs}}(\Lambda', Q)$*.*

Source record: P2-T3 (CW-0470); status PROVED; treatment FULL.

Proof. ($\Rightarrow$) If c is a cell of $K_{\mathrm{abs}}(\Lambda', Q)$ its vertices are in $F(\Lambda', Q)$ and, by Lemma 2.4, $\mathrm{Gen}(c)$ is the set of edge vectors at its smallest vertex (up to sign), which lie in $\mathrm{Gr}_q(\Lambda')$ as generators of a cell of Λ'. ($\Leftarrow$) Let m be the smallest vertex of c. Then $\Pi(m; \mathrm{Gen}(c))$ has independent generators in $\mathrm{Gr}_q(\Lambda')$ and all corners in $V(c) \subseteq F(\Lambda', Q)$, so it is a cell of Λ'; it has vertex set $V(c)$, hence equals c. (1)–(3): for $\Lambda \subseteq \Lambda'$ the vertex condition is automatic; a cell with j independent generators in S needs $j \le \mathrm{rank}\langle S\rangle$; (3) is (1) read for all cells. □

This is the general form of the survival rule (ER-15): the earlier statement that a cell survives iff its generators lie in $\mathrm{Gr} \cap \mathrm{Gr}'$ is correct only when its vertices already lie in F', as they do for $\Lambda \subseteq \Lambda'$.

4.2 Graver sets of $\mathbb{Z}^k$ and of the even-sum lattice

Proposition 4.4. *For $k \ge 2$:* $\mathrm{Gr}(\mathbb{Z}^k) = \{\pm e_i\}$ *and* $\mathrm{Gr}(E_k) = \{\pm 2e_i\} \cup \{\pm e_i \pm e_j : i < j\}$. *The two sets are disjoint.*

Source record: E2-5 (CW-0460); status PROVED; treatment FULL.

Proof. In $\mathbb{Z}^k$ each e_i is minimal, and any nonzero vector dominates some $\pm e_i$. In E_k: $e_i \notin E_k$, so $2e_i$ and $e_i \pm e_j$ have no proper nonzero conformal lattice subvector (the only candidates are $\pm e_i, \pm e_j$), hence are minimal. Conversely let $0 \ne \lambda \in E_k$. If $|\lambda_i| \ge 2$ for some i and $\lambda \ne \pm 2e_i$, then $2\,\mathrm{sign}(\lambda_i)e_i \prec \lambda$ is a lattice vector, so λ is not minimal. If all $|\lambda_i| \le 1$, the support has even size ≥ 2; if it has ≥ 4 elements, then $\mathrm{sign}(\lambda_i)e_i + \mathrm{sign}(\lambda_j)e_j \prec \lambda$ for two support indices. □

(The set $\mathrm{Gr}(E_k)$ is the root system D_k together with $\pm 2e_i$.)

4.3 The three-stage orbit

Theorem 4.5 (three-stage orbit; all k, all admissible prime sets)**.** *Let $k \ge 2$, $p_1 = 2 < p_2 < \cdots < p_k$ primes with $p_i \equiv 2 \pmod 3$, $P = \prod p_i$, $x_0 = P^4/3$. Then $L(x_0) = P^2$, $G(x_0) = 3$, $x_1 = \mathcal{C}(x_0) = 2P^4/3$, $x_2 = \mathcal{C}(x_1) = P^4$, and*

	x_0	x_1	x_2
reduced denominator	3	3	1
box q	$(2, \dots, 2)$	$(2, \dots, 2)$	$(2, \dots, 2)$
$\lvert F\rvert$	$(3^k - 1)/2$	$(3^k - 1)/2$	$3^k - 1$
relation lattice	E_k	E_k	$\mathbb{Z}^k$
D	D_0	$= D_0$	$\supsetneq D_0$

Source record: E2-1 (CW-0456); status PROVED; treatment FULL.

Proof. $d^2 \mid P^4 \iff d \mid P^2$, so $d = \prod p_i^{\alpha_i}$ with $\alpha \in \{0,1,2\}^k$. Since $p_i \equiv -1 \pmod 3$, $d \equiv (-1)^{\sum \alpha_i}$; thus $D(x_0)$ corresponds to the nonzero even-weight vectors, $(3^k + 1)/2 - 1 = (3^k - 1)/2$ of them. The vectors $2e_i$ (that is, $d = p_i^2$) lie in $D(x_0)$ and every d divides P^2, so $L = P^2$. Every $d - 1$ is divisible by 3, and $4 = 2^2 \in D(x_0)$ has $d - 1 = 3$, so $G = 3$. Hence $\mathcal{C}(x_0) = P^4/3 + P^4/3 = 2P^4/3$. In x_1 the numerator $2P^4$ is prime to 3 and $v_2(2P^4) = 5$, $\lfloor 5/2 \rfloor = 2$; the active primes, the box, and (since $d^2 \mid 2P^4 \iff d \mid P^2$) the set D are unchanged,

so are L, G, and $\mathcal{C}(x_1) = 2P^4/3 + P^4/3 = P^4$. For $x_2 = P^4$ the denominator is 1, $D(x_2)$ is the set of all divisors ≥ 2 of P^2 ($3^k - 1$ of them) and the lattice is $\mathbb{Z}^k$. □

Corollary 4.6 (all positive-dimensional cells are lost)**.** *For the transition $x_1 \to x_2$ of Theorem 4.5 and every $k \geq 2$: all vertices persist, no cell of positive dimension of $K_{\mathrm{abs}}(E_k, Q)$ is a cell of $K_{\mathrm{abs}}(\mathbb{Z}^k, Q)$, although $D(x_1) \subsetneq D(x_2)$. The transition $x_0 \to x_1$ is the identity on (Λ, Q).*

Source record: `E2-6` (`CW-0461`); status PROVED; treatment FULL.

Proof. $E_k \subseteq \mathbb{Z}^k$ gives $F \subseteq F'$. By Proposition 4.4 $S = \mathrm{Gr}(E_k) \cap \mathrm{Gr}(\mathbb{Z}^k) = \emptyset$ (the boxes $[-2, 2]^k$ contain all of both Graver sets), so Theorem 4.3(2) applies. □

The corollary generalises the specimen $P = 110$, $k = 3$, in which $13 \to 13 \to 26$ points; that specimen has the following exact counts, obtained here by independent exact code (`scripts/run_exact_checks.py`).

Proposition 4.7 (specimen $k = 3$, FINITE-EXACT)**.** *For $\Lambda = E_3$ and $\Lambda' = \mathbb{Z}^3$ with $Q = [0, 2]^3$: the f-vectors of K_{abs} are $(13, 45, 36, 3)$ and $(26, 51, 33, 7)$; the Betti numbers over $\mathbb{Q}$ are $(1, 6, 6, 0)$ and $(1, 0, 0, 0)$; of the source cells 0 edges, 0 squares, 0 cubes persist, with $45, 36, 3$ lost and $51, 33, 7$ new.*

Source record: `E2-10` (`CW-0464`); status FINITE-EXACT; treatment FINITE.

These are counts for one specimen ($k = 3$, box $[0, 2]^3$), obtained by exhaustive enumeration, with $\partial\partial = 0$ checked. The Betti numbers of the realised union U ($(1, 0, 18, 0)$ for the source and $(1, 0, 0, 0)$ for the target) and the statement that $K_{\mathrm{abs}}(E_3, Q)$ is not embedded are reported by the release from an exact nerve computation and are cited, not re-derived here; the realised $b_2 = 18$ differs from the abstract $b_2 = 6$, which is why every topological statement must name its object.

4.4 A planar family with exact edge counts

Theorem 4.8. *Let $x_h = 21^{2h}/5$, $h \geq 1$. (i) $D(x_h) = \{3^a 7^b : a \equiv b\ (4),\ 0 \leq a, b \leq h,\ (a, b) \neq 0\}$, $L = 21^h$, $G = 20$, $\mathcal{C}(x_h) = x'_h = 21^{2h}/4$; the primes $(3, 7)$ and the box (h, h) are unchanged. (ii) $\Lambda = \{a \equiv b\ (4)\} \subsetneq \Lambda' = \{a \equiv b\ (2)\}$. (iii) $\mathrm{Gr}(\Lambda) = \pm\{(1, 1), (4, 0), (0, 4), (1, -3), (3, -1), (2, -2)\}$, $\mathrm{Gr}(\Lambda') = \pm\{(1, 1), (1, -1), (2, 0), (0, 2)\}$, and $\mathrm{Gr}_h(\Lambda) \cap \mathrm{Gr}_h(\Lambda') = \{\pm(1, 1)\}$ for every $h \geq 1$. (iv) All vertices persist; the persisting edges are exactly the edges $\{u, u + (1, 1)\}$; no 2-cell persists; their number is*

$$E_{\mathrm{kept}}(h) = \sum_{r=0}^{3} c_r(h)^2 - 1, \qquad c_r(h) = \#\{n \in [0, h-1] : n \equiv r\ (4)\}.$$

Source record: `P2-TF1` (`CW-0471`); status PROVED; treatment FULL.

Proof. (i) $d^2 \mid 21^{2h}$ iff $d \mid 21^h$, $d = 3^a 7^b$. Modulo 5: $7 \equiv 2 \equiv 3^3$ and 3 has order 4, so $3^a 7^b \equiv 3^{a+3b} \equiv 3^{a-b}$, which is 1 iff $a \equiv b\ (4)$. Then $(h, h) \in D$, so $L = 21^h$. Every such d has $a + b$ even; since $3 \equiv 7 \equiv -1\ (\mathrm{mod}\ 4)$ we get $d \equiv 1\ (\mathrm{mod}\ 4)$, with $5 \mid d - 1$: $20 \mid G$. And $21 \in D$ ($a = b = 1$) has $21 - 1 = 20$, so $G = 20$. Therefore $\mathcal{C}(x_h) = 21^{2h}(\frac{1}{5} + \frac{1}{20}) = 21^{2h}/4$; the numerator is unchanged, so primes and box are unchanged. (ii) Modulo 4: $3^a 7^b \equiv (-1)^{a+b}$; modulo 5 as

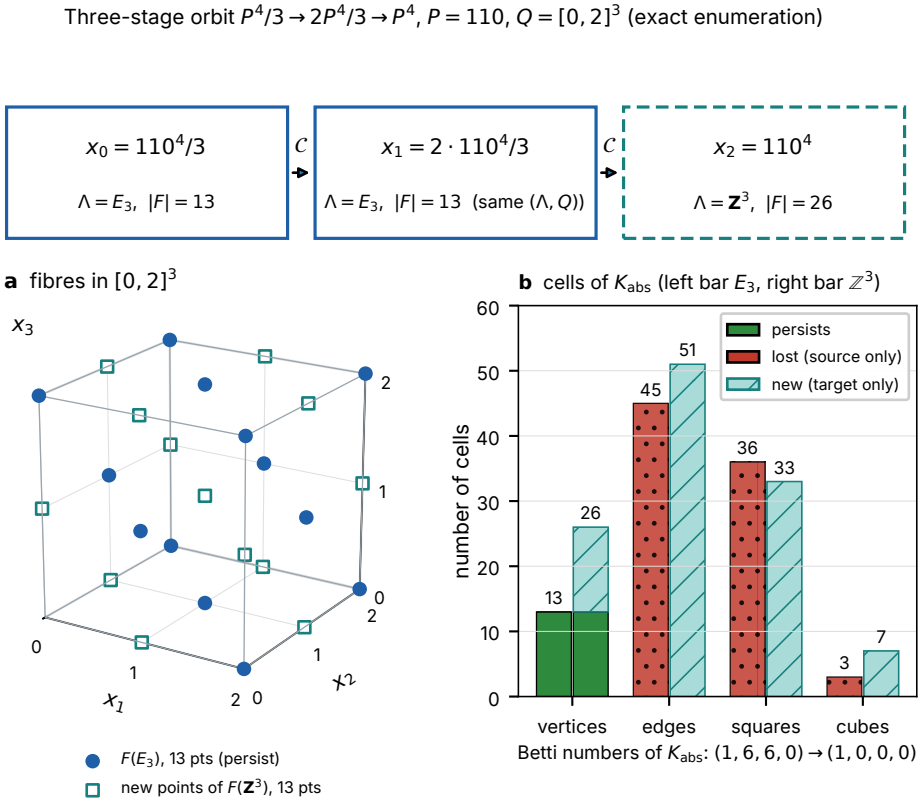


Figure 3: The three-stage orbit $P^4/3 \to 2P^4/3 \to P^4$ for $P = 110$, $k = 3$, $Q = [0, 2]^3$ (Theorem 4.5, Corollary 4.6, and Proposition 4.7; *exact computation*). The first step is the identity on $(\Lambda, Q) = (E_3, Q)$; the second enlarges E_3 to $\mathbb{Z}^3$ ($13 \to 13 \to 26$ occurrences). (a) Fibres: the 13 points of $F(E_3, Q)$ persist, 13 new points of $F(\mathbb{Z}^3, Q)$ appear. (b) Cell counts of the abstract complexes $K_{\text{abs}}(E_3, Q)$, f-vector $(13, 45, 36, 3)$, and $K_{\text{abs}}(\mathbb{Z}^3, Q)$, f-vector $(26, 51, 33, 7)$: all 13 vertices persist (green) and no positive-dimensional cell does (red: lost; hatched teal: new). The abstract Betti numbers change from $(1, 6, 6, 0)$ to $(1, 0, 0, 0)$. Only abstract complexes are counted; the realised unions are compared in Section 6.

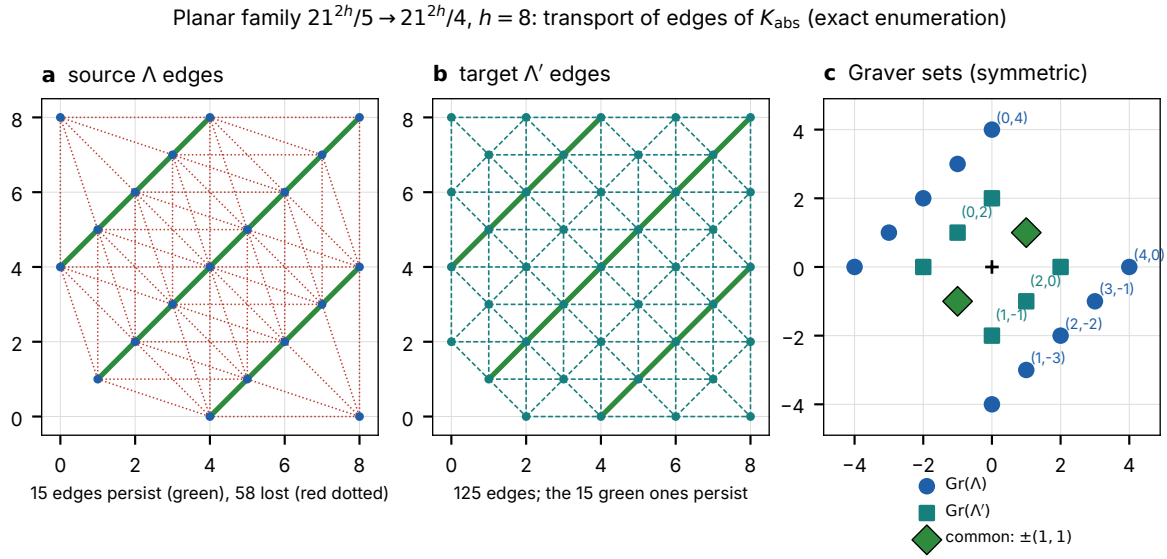


Figure 4: Planar family $21^{2h}/5 \to 21^{2h}/4$ for $h = 8$ (Theorem 4.8; *exact computation* of K_{abs}). (a) The 73 edges of $K_{\mathrm{abs}}(\Lambda, Q)$, $\Lambda = \{a \equiv b\ (4)\}$: the $15 = E_{\mathrm{kept}}(8)$ edges with generator $\pm(1,1)$ persist (green, thick) and the other 58 are lost (red, dotted). (b) The 125 edges of $K_{\mathrm{abs}}(\Lambda', Q)$, $\Lambda' = \{a \equiv b\ (2)\}$, with the persisting edges again green. (c) The Graver sets $\mathrm{Gr}(\Lambda)$ and $\mathrm{Gr}(\Lambda')$ (each symmetric under $v \mapsto -v$) in the window $[-4,4]^2$; they share only $\pm(1,1)$, which is why only $(1,1)$-edges can persist (Theorem 4.3). The panels show abstract complexes (edges only), not the realised unions.

in (i). (iii) Let $v = (a,b) \neq 0$ in Λ. If $a, b \geq 1$ then $(1,1) \preceq v$; if $b = 0$ then $4 \mid a$ and $(4,0) \preceq v$; likewise $a = 0$. If $a > 0 > b$ put $c = -b > 0$, so $a + c \equiv 0\ (4)$; $a \geq 4$ gives $(4,0) \preceq v$ and $c \geq 4$ gives $(0,-4) \preceq v$, so minimal vectors have $a, c \leq 3$ and $a + c = 4$: $(1,-3), (2,-2), (3,-1)$, each minimal because every nonzero proper conformal subvector has $a' + c' \in \{1,2,3\}$. For Λ': this is $\mathrm{Gr}(E_2)$ by Proposition 4.4. The source vectors $(4,0), (0,4), (1,-3), (3,-1), (2,-2)$ dominate $(2,0), (0,2), (1,-1), (1,-1), (1,-1)$ respectively, all in Λ', so they are not target Graver vectors; $(1,1)$ is common; the target vectors $(1,-1), (2,0), (0,2)$ are not in Λ. Region sets are the intersections with $[-h,h]^2$ (Lemma 4.1); $(1,1)$ lies in the box for $h \geq 1$. (iv) $\Lambda \subseteq \Lambda'$ gives $F \subseteq F'$. By Theorem 4.3 an edge persists iff its generator is in $S = \{\pm(1,1)\}$, and a 2-cell needs two independent generators in S: impossible. The $(1,1)$-edges are $\{u, u + (1,1)\}$ with $u = (a,b)$, $a \equiv b\ (4)$, $0 \leq a, b \leq h - 1$, $u \neq 0$; counting residue classes gives $\sum c_r^2 - 1$. □

Remark 4.9. The release also gives closed quasi-polynomial formulas for the number of source and target edges and the limiting survival ratio $1/6$; the quasi-polynomials for the source and target edge counts are valid only for $h \geq 4$ (ER-17), whereas $E_{\mathrm{kept}}(h)$ above holds for all $h \geq 1$. Those counting formulas are not reproduced here. For $h = 1, \ldots, 12$ the value E_{kept}, the equality $S = \{\pm(1,1)\}$ and the non-persistence of 2-cells were confirmed by direct exact enumeration (`scripts/run_exact_checks.py`); this confirms the proof and is not a substitute for it. Figure 4 shows $h = 8$.

5 Rectangular targets and the corner cube

The simplest target is rectangular: $\Lambda' = \bigoplus_i o_i\mathbb{Z}$, which has Graver set $\{\pm o_i e_i\}$. This includes $\Lambda' = \mathbb{Z}^k$ (all $o_i = 1$), the case $b' = 1$. For rectangular targets the realised union has an explicit complement and the transport question becomes exact.

5.1 Gap reduction

Lemma 5.1 (coverage-gap reduction)**.** *Let $F \subseteq F'$ and let $H = \operatorname{conv} F \subseteq H' = \operatorname{conv} F'$. Put $\mathcal{G}' = H' \setminus U'$. Then $U \setminus U' = U \cap \mathcal{G}'$. Hence $U \subseteq U' \iff U \cap \mathcal{G}' = \emptyset$, and $U' = H'$ implies $U \subseteq U'$.*

Source record: `P3-T0` (`CW-0476`); status PROVED; treatment FULL.

Proof. Every cell is the convex hull of its vertices, so $U \subseteq H \subseteq H'$. A point of U outside U' therefore lies in $H' \setminus U'$, and conversely. □

5.2 The corner-cube criterion

Fix a rectangular target $\Lambda' = \bigoplus o_i\mathbb{Z}$, a source Λ with $F(\Lambda, Q) \subseteq F(\Lambda', Q)$, and assume $\tilde{Q}_i = \lfloor q_i/o_i \rfloor \geq 1$ for all i (if some $\tilde{Q}_i = 0$, the coordinate carries no point of $F' \supseteq F$ and the problem is the same problem in $k-1$ coordinates). Put $A_i = o_i\tilde{Q}_i$, $B = \prod[0, A_i]$ and let $O = \{z \in \mathbb{R}^k : z_i < o_i \ \forall i\}$ be the *open corner cube*.

Theorem 5.2 (corner-cube criterion)**.** *(a) $U' = B \setminus O = \{z \in B : \max_i z_i/o_i \geq 1\}$. (b) $U \subseteq U' \iff U \cap O = \emptyset \iff$ no cell of $K_{\mathrm{abs}}(\Lambda, Q)$ meets O. (c) If $U \cap O \neq \emptyset$, the scaled lattice $\tilde{\Lambda} = \operatorname{diag}(o)^{-1}\Lambda$ contains a nonzero vector $v \geq 0$ with $|v|_1 \leq k-1$. (d) No cell of dimension ≤ 1 meets O, in any dimension k. (e) If $k = 2$ then $U \subseteq U'$ (for every rectangular target, including $\mathbb{Z}^2$). (f) If a k-cell meets O, then one of its $(k-1)$-faces (itself a cell) meets O. Hence for $k = 3$ every obstruction is carried by a 2-cell.*

Source record: `P3-T3` (`CW-0479`); status PROVED; treatment FULL.

Proof. (a) The Graver vectors of Λ' in $[-q, q]$ are $\pm o_i e_i$, so the cells of $K_{\mathrm{abs}}(\Lambda', Q)$ are the faces $\prod I_i$ of the grid cubes $\prod[n_i o_i, (n_i+1)o_i]$ $(0 \leq n_i < \tilde{Q}_i)$, with I_i either an endpoint or the interval, all of whose corners are nonzero. ($\supseteq$) Let $z \in B$ with $z_{i_0} \geq o_{i_0}$ for some i_0. Put $I_i = \{z_i\}$ if $z_i/o_i \in \mathbb{Z}$ and $I_i = [n_i o_i, (n_i+1)o_i]$ with $n_i = \lfloor z_i/o_i \rfloor$ otherwise (then $z_i < A_i$, so $(n_i+1)o_i \leq A_i$). Then $\Phi = \prod I_i$ is a grid face containing z whose every vertex has i_0-th coordinate $\geq o_{i_0} > 0$ (if I_{i_0} is an interval then $n_{i_0} \geq 1$; otherwise $I_{i_0} = \{z_{i_0}\}$); so 0 is not a vertex of Φ, Φ is a cell, and $z \in U'$. ($\subseteq$) A cell is a grid face $\prod I_i$ that does not have 0 as a vertex; 0 is a vertex iff every I_i has lower endpoint 0, so some $I_i \subseteq [o_i, \infty)$ and every point of the cell has $z_i \geq o_i$.

(b) $U \subseteq H \subseteq \operatorname{conv} F' \subseteq B$, so $z \in U \setminus U'$ iff $z \in U \cap O$ by (a).

(c) Let $z \in U \cap O$ lie in a cell with vertices $v^{(j)} \in F \subseteq \Lambda$ and $z = \sum \lambda_j v^{(j)}$ (convex). In scaled coordinates $\tilde{v}^{(j)} \in \mathbb{N}^k \setminus \{0\}$ and $\sum_i \tilde{z}_i < k$ because each $\tilde{z}_i < 1$, while $\sum_i \tilde{z}_i = \sum_j \lambda_j |\tilde{v}^{(j)}|_1$. So some vertex has $|\tilde{v}|_1 < k$.

(d) A 0-cell is a nonzero point of $\mathbb{N}^k$ in scaled coordinates, hence has a coordinate ≥ 1 and is not in O. Let $[P, R]$ be an edge, $P, R \in F$, $g = R - P \in \mathrm{Gr}(\Lambda)$ (either orientation), and

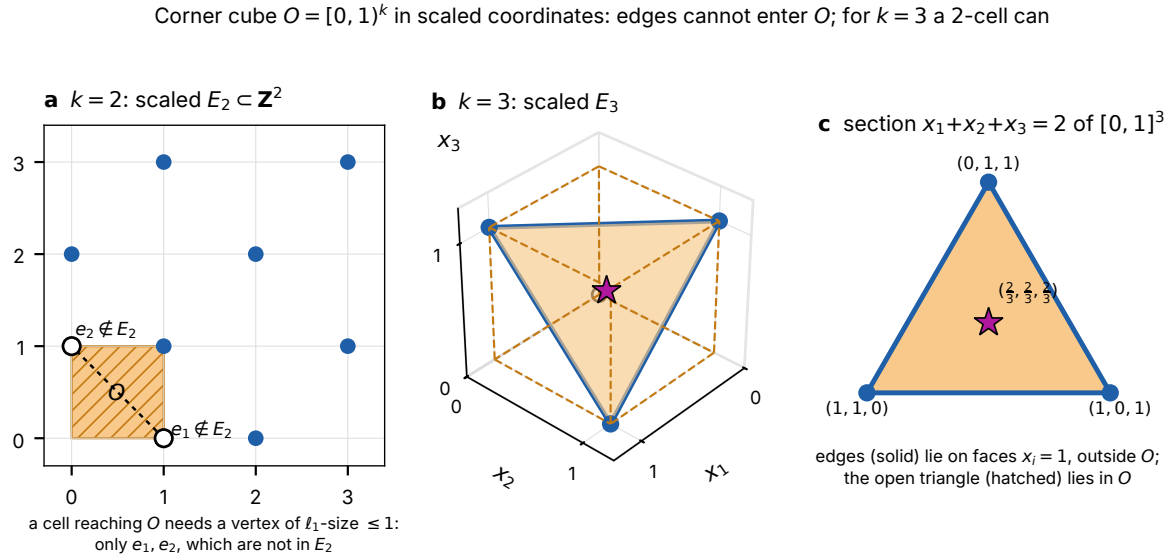


Figure 5: Why dimensions two and three differ (Theorem 5.2(c)–(f), remark 5.3; scaled coordinates; *theorem-level construction with exact coordinates*). (a) $k = 2$, scaled source $E_2 \subset \mathbb{Z}^2$: a cell reaching the corner square $O = [0,1)^2$ would need a nonzero vertex of ℓ_1-size ≤ 1, i.e. e_1 or e_2, neither of which lies in E_2. (b) $k = 3$, scaled source $E_3 \subset \mathbb{Z}^3$: the points $(1,1,0), (1,0,1), (0,1,1)$ of ℓ_1-size 2 *do* lie in E_3; their triangle has its edges on faces $x_i = 1$ of the unit cube, hence outside O, but its relative interior lies in O. The star is the centroid $(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$. (c) The same triangle in the section $x_1 + x_2 + x_3 = 2$. The triangle is not a cell by itself; in Figure 6 it is part of the 2-cell of $K_{\mathrm{abs}}(E_3, Q)$ that realises the obstruction.

suppose $z = (1-t)P + tR \in O$. In scaled coordinates $\tilde{z}_i < 1$ forces $\tilde{P}_i = 0$ or $\tilde{R}_i = 0$ for each i (if both were ≥ 1, then $\tilde{z}_i \geq 1$): the supports are disjoint. Then $\tilde{g} = \tilde{R} - \tilde{P}$ agrees with $\tilde{R}$ on $\operatorname{supp} \tilde{R}$ and is ≤ 0 where $\tilde{P} \neq 0$, so $R \preceq g$, $R \in \Lambda \setminus \{0\}$, and $R \neq g$ because $P \neq 0$. Thus g is not conformally minimal, a contradiction.

(e),(f) Let C be a k-cell with $C \cap O \neq \emptyset$ and suppose $\partial C \cap O = \emptyset$. The cube O is connected and $O = (O \cap \operatorname{int} C) \sqcup (O \setminus C)$ with both parts open, the first nonempty; so $O \subseteq \operatorname{int} C$ and $0 \in \overline{O} \subseteq C$. All vertices of C lie in the closed orthant, hence so does C; 0 is an extreme point of the orthant, so 0 is an extreme point of C, i.e. a vertex, but vertices lie in $F \not\ni 0$. This contradiction shows $\partial C \cap O \neq \emptyset$; ∂C is the union of the $(k-1)$-faces, which are cells. For $k = 2$ these faces are edges, which miss O by (d); cells of dimension ≤ 1 miss O by (d); so $U \cap O = \emptyset$ and (b) gives $U \subseteq U'$. □

Remark 5.3 (why dimension two is special). By (c) a failure needs a nonzero vertex of ℓ_1-size at most $k-1$ in scaled coordinates. For $k = 2$ this means a scaled vertex e_i, a configuration already excluded by (d)–(e); for $k = 3$ the size 2 is reached by vectors such as $(1,1,0)$, which exist in the even-sum lattice, and by (f) the cells that obstruct are squares. This is the first place where dimension enters; the planar hard-cell mechanism of Section 7 is its non-rectangular analogue.

5.3 The even-sum source

Theorem 5.4 (parametric obstruction for the even-sum family)**.** *Let $k = 3$, let Λ' be rectangular and let the scaled source lattice be E_3 (index 2), box $\tilde{Q} \geq (1,1,1)$. Then $U \not\subseteq U'$ iff $\max_i \tilde{Q}_i \geq 2$. The smallest failing scaled box is $(1,1,2)$ up to permutation.*

Source record: `P3-T3(h)` (`CW-0480`); status PROVED; treatment FULL.

Proof. Both scalings by $\mathrm{diag}(o)^{-1}$ preserve the conformal order, Graver sets, cells, and the relation $U \subseteq U'$, so we work in scaled coordinates: $\Lambda = E_3$, $\Lambda' = \mathbb{Z}^3$, box $\tilde{Q}$. ($\Leftarrow$) Say $\tilde{Q}_3 \geq 2$. By Proposition 4.4, $g = (0,1,-1)$ and $h = (1,0,-1)$ are Graver vectors of E_3. The points $(0,0,2), (0,1,1), (1,0,1), (1,1,0)$ have even sum, are nonzero and lie in the box, so $\Pi((0,0,2); g, h)$ is a 2-cell. Its point $(0,0,2) + \frac{2}{3}g + \frac{2}{3}h = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ lies in O, so $U \not\subseteq U'$ by Theorem 5.2(b). ($\Rightarrow$) If all $\tilde{Q}_i = 1$, then $F = \{(1,1,0), (1,0,1), (0,1,1)\}$ ($(1,1,1)$ has odd sum): three points, so there is no 2-cell; U consists of points and edges, which miss O by Theorem 5.2(d). □

The cell and the point are drawn in Figure 6. The first specimen of this phenomenon is the transition $13 \to 26$ of Section 4.3, whose source cell $\Pi((0,0,2); (0,1,-1), (1,0,-1))$ is exactly this one.

5.4 The target union for $b' = 1$

Corollary 5.5 ($U(\mathbb{Z}^k, Q)$)**.** *For $\Lambda' = \mathbb{Z}^k$ and $q \geq (1, \ldots, 1)$: $U(\mathbb{Z}^k, Q) = \{z \in Q : \max_i z_i \geq 1\}$, a star-shaped (hence contractible) set. A convex polytope $\Pi \subseteq Q$ lies in it iff $\Pi \cap [0,1)^k = \emptyset$.*

Source record: `E2-7` (`CW-0462`); status PROVED; treatment ADAPTED.

Proof. Theorem 5.2(a) with all $o_i = 1$; star-shapedness about q: if $\max z_i \geq 1$ then $\max(tz + (1-t)q)_i \geq 1$ for $t \in [0,1]$ since $q_i \geq 1$. □

Remark 5.6 (predictor). For rectangular targets the criterion (b) can be evaluated cell by cell with one linear program per maximal source cell, never building the target complex. The release implements this; correctness is the content of (b). The procedure is an algorithm, not needed for the argument, and is not part of the theorem corpus of this manuscript (`CW-0481` is excluded for that reason).

Source record: `P3-T4` (`CW-0481`); status PROVED; treatment EXCLUDED.

6 Realised transport: what abstract loss does and does not mean

Corollary 4.6 says that the abstract complex of the three-stage orbit loses every positive-dimensional cell. This section shows that the realised union behaves differently: edges are always retained, some squares are not.

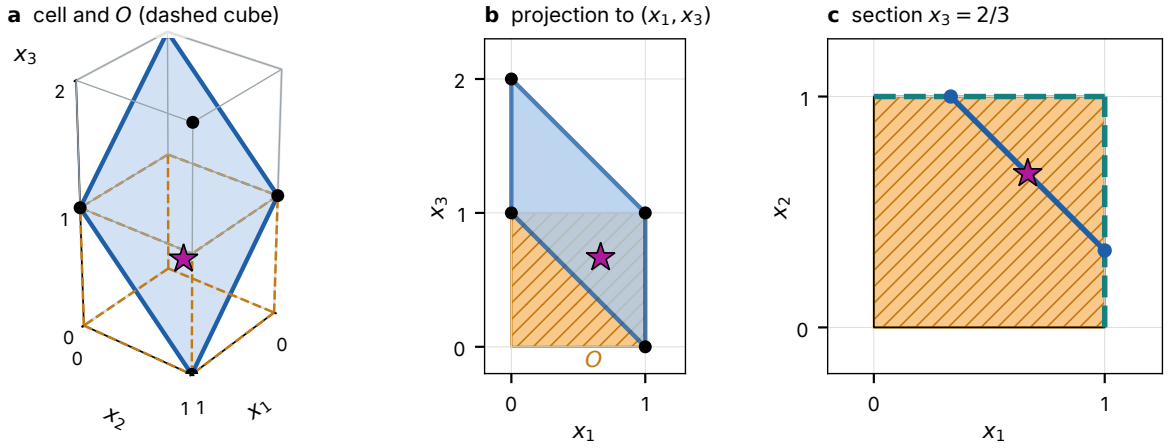


Figure 6: The exact corner-cube witness (Theorem 5.4; scaled $\Lambda = E_3 \subset \Lambda' = \mathbb{Z}^3$, box $(1, 1, 2)$; *exact computation*). (a) The 2-cell $\Pi((0, 0, 2); (0, 1, -1), (1, 0, -1))$ of $K_{\mathrm{abs}}(E_3, Q)$ (a realised parallelogram), the five points of the fibre $F(E_3, Q)$ and the corner cube $O = [0, 1)^3$ (dashed); the star is the witness $(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}) \in O$. (b) Orthogonal projection to (x_1, x_3), complementing the perspective view. (c) The section $x_3 = \frac{2}{3}$: the section of the cell (blue segment from $(1, \frac{1}{3})$ to $(\frac{1}{3}, 1)$) has its interior in the open square O, while the target union $U(\mathbb{Z}^3, Q)$ at this height consists only of the lines $x_1 = 1$, $x_2 = 1$ (teal, dashed), met by the segment only at its end points. Hence the witness is not in $U(\mathbb{Z}^3, Q)$. All three panels show realised objects; none shows K_{emb}.

6.1 Lost edges are geometrically retained

Theorem 6.1. *Let $\Lambda \subseteq \mathbb{Z}^k$ be any lattice and Q a box with all $q_i \geq 1$. Every cell of dimension ≤ 1 of $K_{\mathrm{abs}}(\Lambda, Q)$ lies in $U(\mathbb{Z}^k, Q)$. In particular, for the even-sum lattice E_k every edge segment $[u, u + g]$, $g \in \mathrm{Gr}(E_k)$, of $K_{\mathrm{abs}}(E_k, Q)$ lies in $U(\mathbb{Z}^k, Q)$, although (Corollary 4.6) none of these edges is an edge of $K_{\mathrm{abs}}(\mathbb{Z}^k, Q)$.*

Source record: `E2-8` (`CW-0463`); status PROVED; treatment FULL.

Proof. $\mathbb{Z}^k$ is a rectangular target with all $o_i = 1$ and $F(\Lambda, Q) \subseteq F(\mathbb{Z}^k, Q)$. By Theorem 5.2(d) no cell of dimension ≤ 1 meets $O = [0, 1)^k$, and by Theorem 5.2(a)–(b) a subset of $B = Q$ that misses O lies in $U(\mathbb{Z}^k, Q)$. □

The release states and proves this for E_k by a direct coordinate argument; the form above is the corner-cube form of the same fact. For the specimen $k = 3$, all 45 lost edges were also tested at five interior points each by independent code (a check of the theorem, not an additional proof).

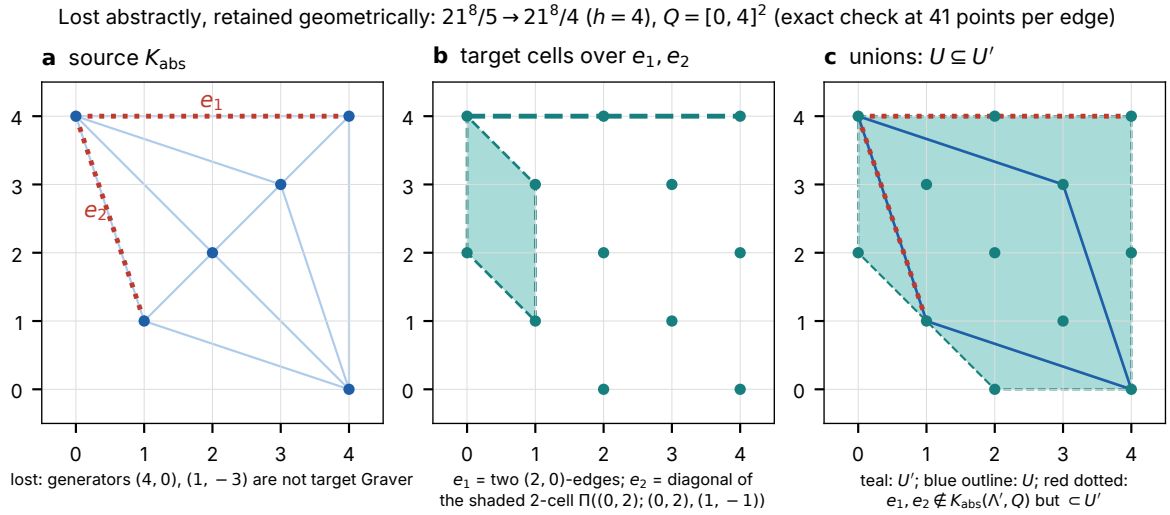


Figure 7: Abstract loss versus realised containment (Theorem 6.1 and Theorem 6.4(L3); planar specimen $21^8/5 \to 21^8/4$, $h = 4$, $Q = [0, 4]^2$; *exact computation*, each edge checked exactly at 41 points). (a) Two edges e_1 (from $(0, 4)$ to $(4, 4)$, generator $(4, 0)$) and e_2 (from $(0, 4)$ to $(1, 1)$, generator $(1, -3)$) of $K_{\text{abs}}(\Lambda, Q)$, $\Lambda = \{a \equiv b \ (4)\}$; neither generator is Graver for $\Lambda' = \{a \equiv b \ (2)\}$, so both edges are lost in $K_{\text{abs}}(\Lambda', Q)$. (b) In the target complex e_1 is the union of two $(2, 0)$-edges and e_2 is a diagonal of the shaded 2-cell $\Pi((0, 2); (0, 2), (1, -1))$. (c) The unions: $U(\Lambda', Q)$ (teal) contains $U(\Lambda, Q)$ (blue outline), and the lost edges (red, dotted) lie inside it. Distinct objects: lost in $K_{\text{abs}}(\Lambda', Q)$, contained in $U(\Lambda', Q)$.

6.2 Cells that leave the union

Proposition 6.2. *Let* $k = 3$, $Q = [0, 2]^3$. *The parallelogram* $c_* = \Pi((0, 0, 2); (0, 1, -1), (1, 0, -1))$ *is a 2-cell of* $K_{\text{abs}}(E_3, Q)$ *(its corners* $(0, 0, 2), (0, 1, 1), (1, 0, 1), (1, 1, 0)$ *lie in* $F(E_3, Q)$ *and its generators are Graver). It contains* $(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}) = (0, 0, 2) + \frac{2}{3}(0, 1, -1) + \frac{2}{3}(1, 0, -1) \in [0, 1)^3$. *Hence* $U(E_3, Q) \not\subseteq U(\mathbb{Z}^3, Q)$ *although* $F(E_3, Q) \subseteq F(\mathbb{Z}^3, Q)$. *Conversely* $(1, 0, 0) \in U(\mathbb{Z}^3, Q) \setminus U(E_3, Q)$.

Source record: `E2-10` (`CW-0464`); status FINITE-EXACT; treatment FINITE.

Proof. The first part is Theorem 5.4 for $\tilde{Q} = (2, 2, 2)$ and Corollary 5.5. For the converse, every nonzero point of E_3 has coordinate sum ≥ 2, hence so has every point of $U(E_3, Q) \subseteq \operatorname{conv} F(E_3, Q)$, while $(1, 0, 0)$ is a point of $F(\mathbb{Z}^3, Q)$. □

(The proposition is carried by the source record E2-9, whose row is the same `CW-0464` area; the classification of *which* source cells leave U' — exactly six, three squares and three cubes, all containing the three points $(0, 1, 1), (1, 0, 1), (1, 1, 0)$ — is a finite exact computation of the release and is cited, not re-derived.)

6.3 Finite findings on topology and specimens

Proposition 6.3 (FINITE-EXACT; scope: the $k = 3$ specimen and two planar specimens)**.** *(a) For the specimen of Proposition 4.7 the Betti numbers of the abstract complexes are*

$(1,6,6,0) \to (1,0,0,0)$ *while the fibre grows; the realised union of the source has Betti numbers* $(1,0,18,0)$ *(nerve computation of the release; cited). (b) For the planar specimen* $21^8/5 \to 21^8/4$ *(*$h = 4$*) all* 8 *abstractly lost edges and the lost square lie in* U'*, with* $U \subsetneq U'$*, realised Betti* $(1,2,0) \to (1,0,0)$ *while the abstract Betti numbers are* $(1,5,0) \to (1,2,6)$*; neither complex is embedded. (c) For the planar specimen* $55^{12}/27 \to 55^{12}/18$ *all* 5 *source edges persist,* U *is embedded with Betti* $(1,0,0)$*, but* U' *is non-embedded with realised Betti* $(1,2,0)$ *(abstract* $(1,8,0)$*): enlargement can create holes.*

Source record: `P2-G` (`CW-0474`); status FINITE-EXACT; treatment FINITE.

Statement (a) first half was recomputed here; the other items are reported by the release and are not re-derived. The planar hole creation and destruction in (b),(c) are explained by frozen planar theory (the source lattices sit at the exceptional hollow box (d,d) or below the stabilisation threshold; ER-19), so they are an observation about transport, not new single-fibre topology.

6.4 Verdict on the plausible laws

Theorem 6.4 (laws refuted by exact witnesses)**.** *None of the following is true in general. (L1)* $D(x) \subseteq D(x')$ *implies persistence of Graver edges. (L2)* $F \subseteq F'$ *implies* $U \subseteq U'$*. (L3) Abstract loss of a cell implies that the cell leaves the target union. (L4) Topology of the abstract complex or of the union is monotone under inclusion of occurrence sets. (L5) A change of denominator changes the relation lattice.*

Source record: `E2-11` (`CW-0465`); status COUNTEREXAMPLE; treatment EXAMPLE.

Proof. (L1): Corollary 4.6. (L2): Proposition 6.2. (L3): Theorem 6.1. (L4): Proposition 4.7 and Proposition 6.3: $b_1(K_{\mathrm{abs}})$ falls from 6 to 0 and $b_2(U)$ from 18 to 0 while F grows from 13 to 26 points (finite-exact specimen). (L5): for the primes $5, 17, 29$ the lattices for $b = 3$ and $b = 12$ coincide (Corollary 3.3); the release's family of 405 transitions of the three-stage type records 276 EQUAL steps with denominator changes of this kind (finite-exact, cited). □

Source record: `E2-P2-V` (`CW-0475`); status PROVED; treatment EXAMPLE.

The release records six preregistered propositions P2-F1–P2-F6, each refuted by an exact witness; among them P2-F4 (abstract edge loss with full geometric containment, 199 corpus rows) is the planar form of (L3) and P2-F5 (the planar pattern extends to $k = 3$) is refuted by the witness of Proposition 6.2. The other verdicts are cited from the release and are not used below.

7 Planar monotonicity

This section proves that in the plane the realised union is monotone under every inclusion of relation lattices and every box. The proof has three parts: a decomposition of cells into smaller cells of the target (Section 7.1), an exact criterion for when a cell can leave the target (Section 7.2), and the geometry of the cells that satisfy that criterion (Sections 7.3 and 7.4). Section 7.5 places the result relative to the inherited planar stabilisation theorem.

7.1 Two tools: conformal decomposition and zonotopal tiling

Lemma 7.1 (conformal decomposition)**.** *Every v in a lattice M is a finite sum of Graver vectors of M, each conformally $\preceq v$.*

Source record: `E2P4v5-L1` (`CW-0499`); status PROVED; treatment FULL.

Proof. Induction on $|v|_1$. For $v = 0$ the sum is empty. Otherwise the set of nonzero lattice vectors $\preceq v$ is finite and nonempty; a member g of minimal ℓ_1-norm is Graver (a nonzero $w' \preceq g$ would also be $\preceq v$ and shorter). Now $v - g \preceq v$ is shorter; apply induction and transitivity of $\preceq$. □

Lemma 7.2 (explicit tiling)**.** *Let $v_1, \dots, v_n$ be nonzero integer vectors spanning a d-dimensional subspace, written in lattice coordinates of the span, and $Z(V) = \sum[0,1]v_i$. Then $Z(V)$ is covered by finitely many tiles T_S, with pairwise disjoint interiors, each a translate of a parallelotope $Z(V_S) = \sum_{i\in S}[0,1]v_i$ with V_S linearly independent, and every vertex of every tile is a subset sum $\sum_{i\in A} v_i$.*

Source record: `E2P4v5-T0` (`CW-0500`); status PROVED; treatment ADAPTED.

Proof. If $n = d$, $Z(V)$ is one parallelotope. Otherwise choose integer heights w_i so generic that for every d-subset S with $\det V_S \neq 0$ and every $i \notin S$ the lifted vector $\tilde{v}_i = (v_i, w_i) \in \mathbb{R}^{d+1}$ is not in $\operatorname{span}(\tilde{v}_s : s \in S)$ (finitely many hyperplanes in w are avoided: the coefficient of w_i is 1). For such S let ν_S be a normal of $\operatorname{span}(\tilde{v}_S)$, signed with negative last coordinate (nonzero: it is $\pm\det V_S$), $A_S = \{i \notin S : \nu_S \cdot \tilde{v}_i > 0\}$ and $T_S = \sum_{i\in A_S} v_i + Z(V_S)$. The face of the lifted zonotope $\tilde{Z} = \sum[0,1]\tilde{v}_i$ on which a functional ν is maximal is $\sum_{\nu\cdot\tilde{v}_i>0} \tilde{v}_i + \sum_{\nu\cdot\tilde{v}_i=0}[0,1]\tilde{v}_i$; by genericity the facets of $\tilde{Z}$ are the sets $\sum_{i\in A_S} \tilde{v}_i + Z(\tilde{v}_S)$, and such a facet is *lower* iff ν_S has negative last coordinate. The lower facets project onto T_S injectively, and *by the lifting lemma (Lemma B.1)* they cover the shadow $Z(V)$ with disjoint interiors; (c) is the definition of T_S. □

The lifting fact. The covering property used above is the standard statement that the lower facets of a lifted polytope project onto its shadow with disjoint interiors (the regular subdivision induced by a lifting, [10, 11]; it is the construction behind the Bohne–Dress correspondence for zonotopal tilings [15, 12, 13, 14]). It is proved in appendix B (Lemma B.1), so Lemma 7.2 is PROVED in full; the qualification of the first version is withdrawn. The release also checks the lemma numerically (volume sum $= \sum_S |\det V_S|$ and point location for random configurations), which is evidence for the implementation, not part of the proof.

7.2 The zero-vertex criterion

Let $\Lambda \subseteq \Lambda' \subseteq \mathbb{Z}^k$ be lattices and Q a box.

Proposition 7.3 (zero-vertex criterion)**.** *Let $c = \Pi(u; g_1, \dots, g_j)$ be a cell of $K_{\mathrm{abs}}(\Lambda, Q)$, and call* zero-vertex decomposition *a choice of $s_i \in \Lambda'$ with $s_i \preceq g_i$ (possibly 0 or g_i) and $u + \sum s_i = 0$.*

(Z.1) If c has no zero-vertex decomposition then $c \subseteq U(\Lambda', Q)$.

(Z.2) c has a zero-vertex decomposition iff $m(c) := u + \sum_i g_i^- = 0$ and $g_i^- \in \Lambda'$ for all i (then $s_i = g_i^-$).

(Z.3) Every cell has 2^j representations (base corner and generator signs), and Z.1 applies to each. Hence $c \subseteq U(\Lambda', Q)$ unless $m(c) = 0$ and $g_i^- \in \Lambda'$ for all i (equivalently $g_i^+ \in \Lambda'$, since $g_i \in \Lambda \subseteq \Lambda'$). Cells with this property are hard.

(Z.4) There are no hard cells of dimension ≤ 1 (edges never obstruct).

Source record: `E2P4v5-Z` (`CW-0501`); status PROVED; treatment FULL.

Source record: `P4-Z1` (`CW-0490`); status PROVED; treatment FULL.

Source record: `P4-Z2` (`CW-0496`); status PROVED; treatment FULL.

Source record: `P4-Z1.a` (`CW-0491`); status PROVED; treatment FULL.

Proof. (Z.1) By Lemma 7.1 write each $g_i = \sum_t h_{i,t}$ with $h_{i,t} \in \mathrm{Gr}(\Lambda')$ and $h_{i,t} \preceq g_i$. Then $Z' = u + \sum_{i,t}[0,1]h_{i,t}$ contains c (for $\lambda_i \in [0,1]$, $\lambda_i g_i = \sum_t \lambda_i h_{i,t}$). Tile Z' by Lemma 7.2. A vertex of a tile is $u + \sum_{(i,t)\in A} h_{i,t} = u + \sum_i s_i$, where s_i is a partial sum of the pieces of g_i; all pieces have the sign pattern of g_i, so $s_i \in \Lambda'$ and $s_i \preceq g_i$. For each coordinate ℓ the vertex coordinate lies between $u_\ell + \sum_i \min(g_{i,\ell}, 0)$ and $u_\ell + \sum_i \max(g_{i,\ell}, 0)$, which are coordinates of two corners of c, hence in $[0, q_\ell]$: the vertex is in Q. It is nonzero by hypothesis. So every tile (generators $h_{i,t}$ Graver in Λ' and in the box, independent, all vertices in $F(\Lambda', Q)$) is a cell of Λ', and $c \subseteq Z' \subseteq U(\Lambda', Q)$.

(Z.2) ($\Rightarrow$) Coordinatewise $s_{i,\ell} \geq \min(g_{i,\ell}, 0)$, so $0 = u_\ell + \sum_i s_{i,\ell} \geq m_\ell$; but m_ℓ is a coordinate of the corner $u + \sum_{g_{i,\ell}<0} g_i$ of c, hence ≥ 0. So equality holds throughout: $m = 0$ and $s_{i,\ell} = \min(g_{i,\ell}, 0)$, that is $s_i = g_i^- \in \Lambda'$. ($\Leftarrow$) Take $s_i = g_i^- \preceq g_i$.

(Z.3) $m(c)$ is the coordinatewise minimum of the corners, independent of the representation; the representation with generators $\epsilon_i g_i$ has outgoing negative parts $(\epsilon_i g_i)^- \in \{g_i^-, -g_i^+\}$. All 2^j representations have a zero-vertex decomposition iff $m(c) = 0$ and both g_i^- and g_i^+ lie in Λ' for every i.

(Z.4) If a cell $\Pi(u; g)$ were hard then $u + g^- = 0$, so $-u = g^-$, a nonzero element of Λ ($u \in F \subseteq \Lambda$) with $g^- \preceq g$; and $g^- \neq g$, else $u + g = 0 \notin F$. This contradicts $g \in \mathrm{Gr}(\Lambda)$. Points are covered trivially. □

The release gives (Z.1) and (Z.2) first as P4-Z1, P4-Z2 and then re-proves them with a complete argument as Z; the three records are merged here with one proof.

7.3 Adjacent corners overlap in support

Lemma 7.4. *If P, P' are corners of a cell of $U(\Lambda, Q)$ that differ by one generator $w = P' - P$, then* $\operatorname{supp} P \cap \operatorname{supp} P' \neq \emptyset$.

Source record: `E2P4v5-LG` (`CW-0502`); status PROVED; treatment FULL.

Proof. P, P' are nonzero and ≥ 0. If the supports were disjoint then $P' \preceq w$ and $-P \preceq w$ (on $\operatorname{supp} P'$, $w_\ell = P'_\ell \geq 0 = P_\ell$; on $\operatorname{supp} P$, $w_\ell = -P_\ell \leq 0 = P'_\ell$; zero elsewhere), both lie in Λ and are nonzero, $P' \neq w$ (else $P = 0$) and $-P \neq w$ (else $P' = 0$): w would not be Graver. □

7.4 The planar theorem

Theorem 7.5 (planar monotonicity)**.** *Let $\Lambda \subseteq \Lambda' \subseteq \mathbb{Z}^2$ be lattices of rank 2 and $Q = [0,a]\times[0,b]$ any box. Then $U(\Lambda, Q) \subseteq U(\Lambda', Q)$.*

Source record: `P5-P1` (`CW-0504`); status PROVED; treatment FULL.

Source record: `P4-P1` (`CW-0497`); status PROVED; treatment FULL.

Proof. Points are trivial and edges are covered by Proposition 7.3(Z.4) and (Z.3). Let c be a 2-cell. If it is not hard, Proposition 7.3(Z.3) applies. Let it be hard. Let $k', l' \geq 1$ be minimal with $k'e_1, l'e_2 \in \Lambda'$ (they exist: finite index), so $\{t : te_1 \in \Lambda'\} = k'\mathbb{Z}$. Then $k'e_1$ and $l'e_2$ are Graver in Λ': a nonzero conformal subvector of $k'e_1$ in Λ' is $(t, 0)$ with $0 < t \leq k'$, so $t = k'$.

(1) $m(c) = 0$ gives a corner A with $x = 0$ and a corner B with $y = 0$; both are nonzero, so $A = (0, y_0)$, $B = (x_0, 0)$ with $x_0, y_0 > 0$.

(2) $\operatorname{supp} A = \{2\}$ and $\operatorname{supp} B = \{1\}$ are disjoint, so by Lemma 7.4 A, B are not adjacent: they are opposite corners. Let D, E be the other two.

(3) Write the edge vectors at A as g', h' with $B = A + g' + h'$, $D = A + g'$, $E = A + h'$. Since $D, E \in Q$: $g'_1, h'_1 \geq 0$ and $y_0 + g'_2, y_0 + h'_2 \geq 0$. As $g'_2 + h'_2 = -y_0$, both $g'_2, h'_2 \leq 0$. Write $g' = (g_1, -g_2)$, $h' = (h_1, -h_2)$ with $g_1, g_2, h_1, h_2 \geq 0$, $g_1 + h_1 = x_0$, $g_2 + h_2 = y_0$; then $D = (g_1, h_2)$, $E = (h_1, g_2)$.

(4) All four numbers are positive: D is adjacent to both A and B and nonzero, so by Lemma 7.4 $\operatorname{supp} D$ meets $\{2\}$ and $\{1\}$: $h_2 > 0$ and $g_1 > 0$. Likewise for E: $h_1 > 0$, $g_2 > 0$.

(5) Hardness: $g'^{+} = (g_1, 0)$, $g'^{-} = (0, -g_2)$, $h'^{+} = (h_1, 0)$, $h'^{-} = (0, -h_2)$ lie in Λ' (up to the sign convention of Proposition 7.3(Z.3)). Hence $k' \mid g_1, h_1$ and $l' \mid g_2, h_2$, so $g_1, h_1 \geq k'$ and $g_2, h_2 \geq l'$; $x_0 = g_1 + h_1 \in k'\mathbb{Z}$, $x_0 \geq 2k'$, $y_0 \in l'\mathbb{Z}$, $y_0 \geq 2l'$.

(6) The four corners satisfy $x/k' + y/l' \geq 2$ (A: $y_0/l' \geq 2$; B: $x_0/k' \geq 2$; D: $g_1/k' + h_2/l' \geq 2$; E likewise). By convexity every $p \in c$ has $p_1/k' + p_2/l' \geq 2$, so $p_1 \geq k'$ **or** $p_2 \geq l'$.

(7) The rectangles $R_{ij} = [ik', (i+1)k'] \times [jl', (j+1)l']$, $0 \leq i < x_0/k'$, $0 \leq j < y_0/l'$, tile $[0, x_0] \times [0, y_0] \subseteq Q$ (the box contains the bounding box of the corners). Their corners lie in $k'\mathbb{Z} \times l'\mathbb{Z} \subseteq \Lambda'$, in Q, and are nonzero except the corner 0 of R_{00}. So each R_{ij}, $(i, j) \neq (0, 0)$, is the cell $\Pi(\text{corner}; k'e_1, l'e_2)$ of $U(\Lambda', Q)$. A point $p \in c$ with $p_1 \geq k'$ lies in some R_{ij} with $i \geq 1$ (because $x_0/k' \geq 2$), and one with $p_2 \geq l'$ lies in some R_{ij} with $j \geq 1$. Hence $c \subseteq U(\Lambda', Q)$. □

Remark 7.6 (why the plane works). In the proof the cell escapes only through the origin, and the escape is blocked by a fact that is special to $k = 2$: opposite axis corners $A = (0, y_0)$, $B = (x_0, 0)$ force every point of the cell to satisfy $x/k' + y/l' \geq 2$, and for two ratios "sum ≥ 2" implies "one ratio ≥ 1". For three ratios a sum of 2 is compatible with all ratios < 1: in Theorem 8.2 the failing cell has all four corners at ratio sum exactly 2 (with axis orders $(2, 2, 2)$), and its point $(\frac{2}{3}, \frac{5}{3}, \frac{5}{3})$ has ratios $(\frac{1}{3}, \frac{5}{6}, \frac{5}{6})$, all < 1. This explains where the planar argument cannot be repeated; it is not by itself a theorem about dimension three, and the failures proved in Section 8 do not depend on it.

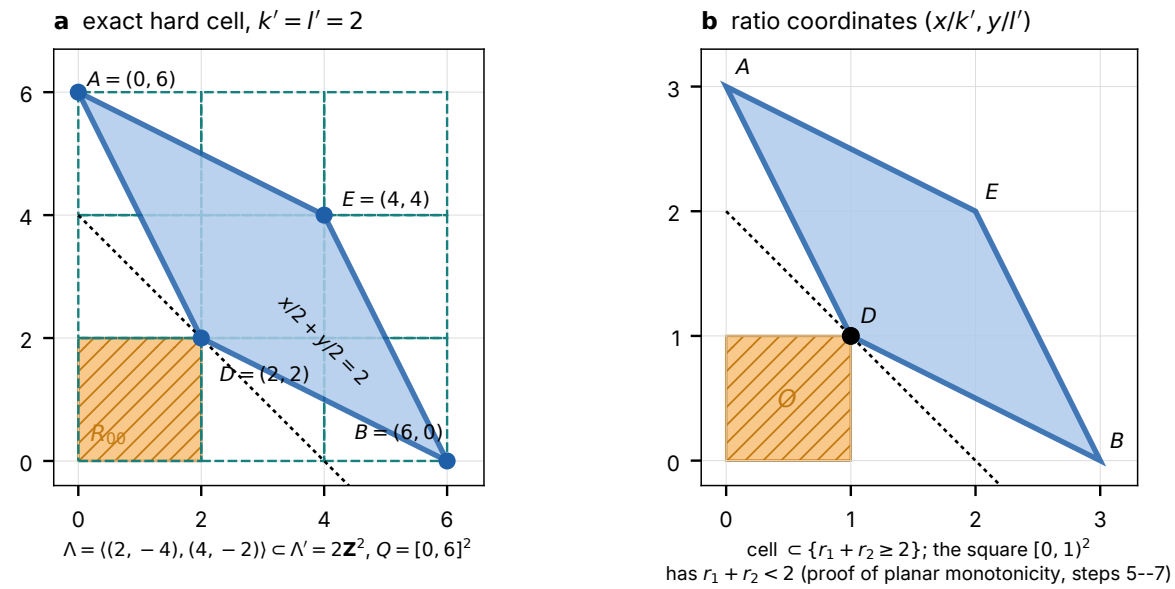


Figure 8: Geometry of the planar proof (Theorem 7.5, steps (1)–(7); *exact computation of one instance of the construction*). Hard cell $c = \Pi((0,6);(2,-4),(4,-2))$ of the source lattice $\{x + 2y \equiv 0,\ 2x + y \equiv 0\ (6)\}$, target $\Lambda' = 2\mathbb{Z}^2$ (so $k' = l' = 2$), box $[0,6]^2$. (a) Opposite axis corners $A = (0,6)$ and $B = (6,0)$ and the other two corners $D = (2,2)$, $E = (4,4)$; the dotted line $x/k' + y/l' = 2$ (i.e. $x + y = 4$) passes through D, the corners A, B, E lie strictly above it, and the whole cell lies on or above it; the 2×2 rectangles of the target tiling of $[0, x_0] \times [0, y_0]$ are dashed teal, and the corner square $[0,2]^2$ at the origin (orange, hatched) is the region the cell never enters. (b) The same cell in the ratio coordinates $(x/k', y/l')$, where all four corners satisfy the inequality; by convexity so does every point of c. All 441 grid points of c were tested exactly against $U(2\mathbb{Z}^2, Q)$. The cell is an actual hard cell, not a schematic; the picture illustrates the general argument by this instance.

Corollary 7.7 (earlier planar conjectures)**.** *Conjecture CP2 (planar containment for $\Lambda \subsetneq \Lambda'$, same primes and box) and conjecture CP3-A (residual planar containment) of the earlier releases are theorems for nested lattices: they are Theorem 7.5. Both records remain in the crosswalk with their historical status OPEN; they are not restated.*

Source record: `CP2` (`CW-0473`); status OPEN; treatment EXCLUDED.

Source record: `CP3-A` (`CW-0482`); status OPEN; treatment EXCLUDED.

7.5 Relation to the inherited planar stabilisation theorem

GEO-I (v004 S1, cited as `CW-0477`) reduces every planar lattice Λ_b to a normal form $\mathrm{diag}(a, c)\,\Gamma_{d,s}$ with $\Gamma_{d,s} = \{(i, j) : j \equiv si \pmod d\}$, s a unit, and box $(\lfloor q_1/a \rfloor, \lfloor q_2/c \rfloor)$. The inherited theorem G12-T8 (GEO-I v012; cited as `CW-0478`, not reproved) states: for $d \geq 2$, normalised box $\geq (d, d)$ and $\neq (d, d)$ one has $U = \mathrm{conv}\, F$. In that regime $U \subseteq U'$ follows from $F \subseteq F'$ by Lemma 5.1 with no hypothesis on the lattices. Theorem 7.5 complements this: it needs $\Lambda \subseteq \Lambda'$ but holds for *every* box, including the exceptional hollow box (d, d) and sub-threshold boxes, where $U \neq \mathrm{conv}\, F$. The planar hole creation and destruction of Proposition 6.3 occurs exactly at those exceptional boxes (ER-19).

Source record: `P3-L1` (`CW-0477`); status PROVED; treatment CITED.

Source record: `P3-C1` (`CW-0478`); status PROVED; treatment CITED.

7.6 Scope for collective transitions

Proposition 7.8 (scope of the FIELD-I corollary (ER-23))**.** *Let $x \to \mathcal{C}(x)$ be an admissible planar transition ($k = 2$, same primes and box). Then $U(\Lambda_x, Q) \subseteq U(\Lambda_{\mathcal{C}(x)}, Q)$ provided $\Lambda_x \subseteq \Lambda_{\mathcal{C}(x)}$. The collective map guarantees only $F \subseteq F'$, and lattices that shrink or are incomparable do occur (Proposition 3.6), so Theorem 7.5 does not by itself settle every planar transition.*

Source record: `P5-P1.a` (`CW-0505`); status PROVED; treatment FULL.

Proof. The first sentence is Theorem 7.5; the rest restates Proposition 3.5(b) and Proposition 3.6. □

For a transition with $F \subseteq F'$ but incomparable lattices, containment of the unions is not established in this manuscript.

8 Dimension three: obstruction to preservation

For $k = 3$ the inclusion $\Lambda \subseteq \Lambda'$ no longer implies $U(\Lambda, Q) \subseteq U(\Lambda', Q)$. Theorem 5.4 already gives an infinite family with rectangular target $\bigoplus o_i\mathbb{Z}$. This section shows that the failure does not depend on the target having a coordinate on which it is "free", and gives the exact structure of the failing cells.

8.1 Dilation invariance

Theorem 8.1 (diagonal dilation)**.** *Let $D = \mathrm{diag}(t_1, \dots, t_k)$ with t_i positive integers and M a lattice. Then $\mathrm{Gr}(DM) = D\ \mathrm{Gr}(M)$, $F(DM, DQ) = D\,F(M, Q)$, cells go to cells, $U(D\Lambda, DQ) = D\,U(\Lambda, Q)$, and*

$$U(\Lambda, Q) \subseteq U(\Lambda', Q) \iff U(D\Lambda, DQ) \subseteq U(D\Lambda', DQ).$$

Source record: `P5-D1` (`CW-0507`); status PROVED; treatment FULL.

Proof. $D : M \to DM$ is a bijection of lattices and $Dw \preceq Dv \iff w \preceq v$ (signs agree and $|t_i w_i| \le |t_i v_i| \iff |w_i| \le |v_i|$), so D preserves the conformal order, hence Graver sets, independence, and the corner condition; it maps $M \cap [0, Q]$ onto $DM \cap [0, DQ]$ and 0 to 0. Cells are affine images of cubes. □

Dilation changes the axis orders: the property "$e_c \notin \Lambda'$" is not dilation invariant, and a failure with a free coordinate becomes a failure in which every axis order is ≥ 2.

8.2 A failure without free coordinate, and its dilation family

Theorem 8.2 (failure with no free coordinate)**.** *Let $E_3 = \{z \in \mathbb{Z}^3 : \sum z_i$ even$\}$, $\Lambda = 2E_3 \subseteq \Lambda' = 2\mathbb{Z}^3$ (axis orders $(2, 2, 2)$: no e_i lies in Λ') and $Q = (2, 4, 2)$. Then $U(\Lambda, Q) \not\subseteq U(\Lambda', Q)$.*

Source record: `P5-C1` (`CW-0508`); status PROVED; treatment FULL.

Proof. $\mathrm{Gr}(2E_3) = 2\ \mathrm{Gr}(E_3)$ (Theorem 8.1 and Proposition 4.4). Let $g = (0, 2, -2) = 2(0, 1, -1)$, $h = (2, -2, 0) = 2(1, -1, 0)$, $u = (0, 2, 2)$. The corners $(0, 2, 2), (0, 4, 0), (2, 0, 2), (2, 2, 0)$ are nonzero points of $2E_3 \cap Q$, so $c = \Pi(u; g, h)$ is a 2-cell of $U(\Lambda, Q)$. The point $p = u + \frac{1}{6}g + \frac{1}{3}h = (\frac{2}{3}, \frac{5}{3}, \frac{5}{3})$ lies in c. In $U(2\mathbb{Z}^3, Q)$ the Graver vectors are $\pm 2e_i$, so every cell is $\Pi(c'; 2e_i, i \in I)$ with $c' \in 2\mathbb{Z}^3$. A cell containing p must, for each coordinate i with $0 < p_i < 2$ (all three do), use the generator $2e_i$ with base coordinate 0; that cell is $[0, 2]^3$, whose corner 0 is excluded. So $p \notin U(\Lambda', Q)$. □

Corollary 8.3 (infinite family)**.** *If (Λ, Λ', Q) fails (i.e. $U(\Lambda, Q) \not\subseteq U(\Lambda', Q)$) then so does every dilation $(D\Lambda, D\Lambda', DQ)$. In particular every failing triple with a free coordinate, such as the scaled $E_3 \subset \mathbb{Z}^3$, $Q = (1, 2, 1)$, $p = (\frac{1}{3}, \frac{5}{6}, \frac{5}{6})$, generates failing triples all of whose axis orders are ≥ 2.*

Source record: `P5-C2` (`CW-0509`); status PROVED; treatment FULL.

Proof. Theorem 8.1. □

Remark 8.4 (relation to the corner-cube theorem). Theorem 8.2 is the instance $o = (2, 2, 2)$ of Theorem 5.4: the scaled source lattice is E_3, the scaled box is $(1, 2, 1)$ with maximum 2, and the scaled point is $(\frac{1}{3}, \frac{5}{6}, \frac{5}{6}) \in O$. The target $2\mathbb{Z}^3$ is rectangular in the sense of Section 5; the release derived the same example independently and emphasises "no free coordinate" because the earlier three-dimensional conjecture was formulated in those terms. (The identification is derived here from the two cited theorems and the scaling invariance; it is an elementary observation.) The recorded 57 lattice-level failures with a free coordinate are finite-exact data of the releases and are not used.

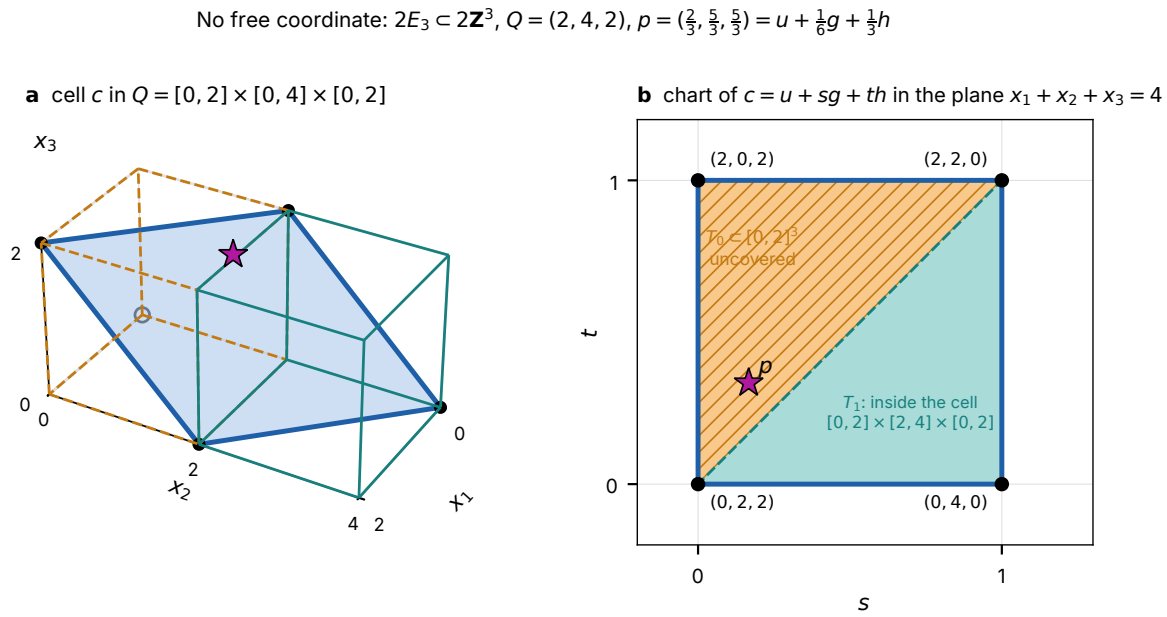


Figure 9: Three-dimensional non-containment (Theorem 8.2; $\Lambda = 2E_3 \subset \Lambda' = 2\mathbb{Z}^3$, $Q = (2, 4, 2)$; *exact computation*). (a) The 2-cell $c = \Pi((0, 2, 2); (0, 2, -2), (2, -2, 0))$ of $K_{\text{abs}}(2E_3, Q)$ in the box $[0, 2] \times [0, 4] \times [0, 2]$ with its four corners $(0, 2, 2)$, $(0, 4, 0)$, $(2, 0, 2)$, $(2, 2, 0)$. (b) The chart $c = u + sg + th$, $(s, t) \in [0, 1]^2$, in the plane $x_1 + x_2 + x_3 = 4$: the part T_0 of c inside the cube $[0, 2]^3$ (light orange) is not covered by any cell of $K_{\text{abs}}(2\mathbb{Z}^3, Q)$, whereas the part T_1 (outside that cube) is. The star is the witness $p = (\frac{2}{3}, \frac{5}{3}, \frac{5}{3}) = u + \frac{1}{6}g + \frac{1}{3}h \in T_0$, in c but in no cell of $K_{\text{abs}}(2\mathbb{Z}^3, Q)$ (exact enumeration); hence $U(2E_3, Q) \not\subseteq U(2\mathbb{Z}^3, Q)$.

8.3 A target that is not split in any coordinate

Proposition 8.5 (FINITE-EXACT; body-centred target)**.** *Let* $\Lambda = 2E_3$, $\Lambda' = \text{BCC} = \{z \in \mathbb{Z}^3 : z_1 \equiv z_2 \equiv z_3 \ (2)\}$ *(index* 4*, axis orders* $(2, 2, 2)$*, not split in any coordinate:* $\Lambda' \neq (\Lambda' \cap \{z_i = 0\}) \oplus (\Lambda' \cap \mathbb{R}e_i)$ *for each* i*) and* $Q = (2, 4, 2)$*. Then* $\Lambda \subseteq \Lambda'$ *and* $U(\Lambda, Q) \not\subseteq U(\Lambda', Q)$*; the cell* c *of Theorem 8.2 contains the point* $(\frac{5}{6}, \frac{5}{3}, \frac{3}{2}) = u + \frac{1}{4}g + \frac{5}{12}h$*, which is not in* $U(\text{BCC}, Q)$*. The same source cell is contained in* $U(E_3, Q)$ *(release; also sampled at* 49 *interior points here).*

Source record: `P5-C3` (`CW-0510`); status FINITE-EXACT; treatment FINITE.

Evidence and scope. The statement is a finite exact computation: for $Q = (2, 4, 2)$, $K_{\text{abs}}(\text{BCC}, Q)$ has 13 vertices, 33 edges, 17 squares and one cube, and the point lies in no cell. The release certified this by three routes (an engine with an independent certificate checker, a mini brute-force implementation, and the preregistered pinned engine); it was reproduced here with a fourth, independent exact implementation (`scripts/run_exact_checks.py`). Non-splitness of BCC: for each i, $\Lambda' \cap \{z_i = 0\} = 2\mathbb{Z}^2$ and $\Lambda' \cap \mathbb{R}e_i = 2\mathbb{Z}e_i$, whose direct sum has index 8 in $\mathbb{Z}^3$, while $[\mathbb{Z}^3 : \Lambda'] = 4$; so Λ' is not that direct sum. The release also records failure for $\Lambda' = E_2 \times 2\mathbb{Z}$ (split in one coordinate), cited. **Consequence:** neither "no free coordinate" nor even "no split coordinate" implies preservation; the 3-dimensional conjecture of the earlier assessment ($\Lambda \subseteq \Lambda' \subseteq \mathbb{Z}^3$, $e_1, e_2, e_3 \notin \Lambda' \Rightarrow U \subseteq U'$) is false, by Theorem 8.2 already.

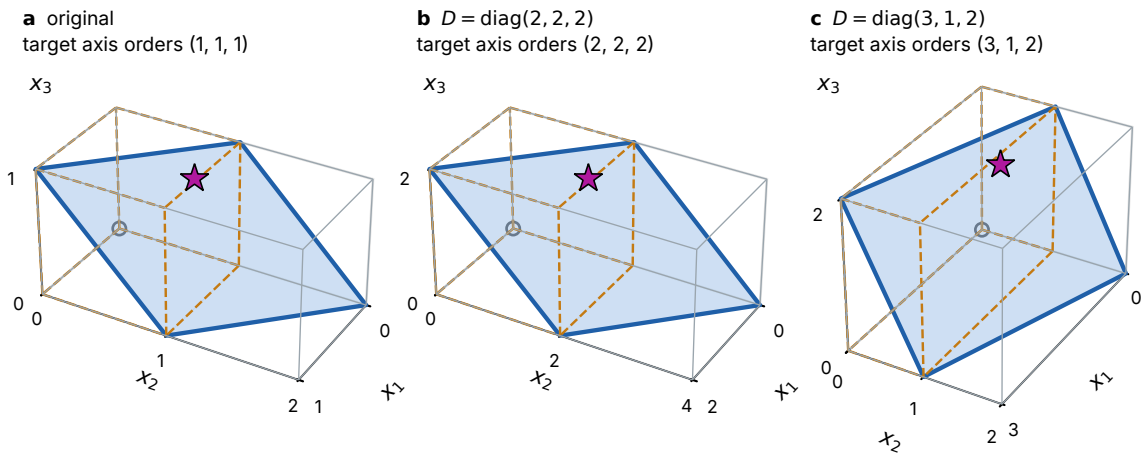


Figure 10: Diagonal dilation preserves failure (Theorem 8.1 and Corollary 8.3; *exact computation*). The failing pair $(E_3 \subset \mathbb{Z}^3, Q = (1,2,1))$ with its witness $(\frac{1}{3}, \frac{5}{6}, \frac{5}{6})$ (a) and two dilations: $D = \mathrm{diag}(2,2,2)$ giving $(2E_3 \subset 2\mathbb{Z}^3, (2,4,2))$ with witness $(\frac{2}{3}, \frac{5}{3}, \frac{5}{3})$ (b) — the configuration of Figure 9 — and $D = \mathrm{diag}(3,1,2)$ giving the box $(3,2,2)$ with witness $(1, \frac{5}{6}, \frac{5}{3})$ (c). In each panel the transformed box, the cell and the transformed corner cube (dashed) are drawn; the target axis orders are $(1,1,1)$, $(2,2,2)$ and $(3,1,2)$, so in (a) every coordinate is free, in (b) none is, and in (c) only x_2 is. Failure was verified exactly in all three cases.

8.4 Cells inside a coordinate plane are preserved

Theorem 8.6. *Let $\Lambda \subseteq \Lambda' \subseteq \mathbb{Z}^3$ and let c be a cell of $U(\Lambda, Q)$ contained in a coordinate plane $H = \{x_i = 0\}$. Then $c \subseteq U(\Lambda', Q)$. So a failing cell is never in a coordinate plane.*

Source record: `P5-F1` (`CW-0506`); status PROVED; treatment FULL.

Proof. A vector conformally below one in H lies in H, so $\mathrm{Gr}(M \cap H) = \mathrm{Gr}(M) \cap H$ for every lattice M. Hence c is a cell of the planar pair $\Lambda \cap H \subseteq \Lambda' \cap H$ (both of rank 2), and Theorem 7.5 gives $c \subseteq U(\Lambda' \cap H, Q \cap H)$. Cells of $\Lambda' \cap H$ with corners in $F(\Lambda' \cap H)$ are cells of Λ' (their generators are Graver in Λ' by the same identity), so $U(\Lambda' \cap H, Q \cap H) \subseteq U(\Lambda', Q)$. □

8.5 Finite census (finite-exact)

Proposition 8.7 (FINITE-EXACT; scope as stated)**.** *Targets of index* 2–6 *without a free coordinate (*96 *lattices), sources of index* 2–5 *(*8,256 *pairs), all hard cells with bounding box* $\leq Q_{\max} = (\min(3k_c, 12))_c$*:* 24,178 *hard cells (*6,825 *squares,* 17,353 *cubes), each tested against its own worst box.* 192 *hard cells fail; the failing targets are* 19 *of the* 96*: all* 12 *split targets and* 7 *of* 84 *non-split targets (BCC and six of index* 6*). The remaining* 77 *non-split targets (including E_3 and every target of index* 3 *and* 5*) have no failure among* 19,794 *hard cells. An extended run (sources of index* 6–9 *in the* 18 *clean targets of index* ≤ 4*) found* 19,877 *hard cells and no failure.*

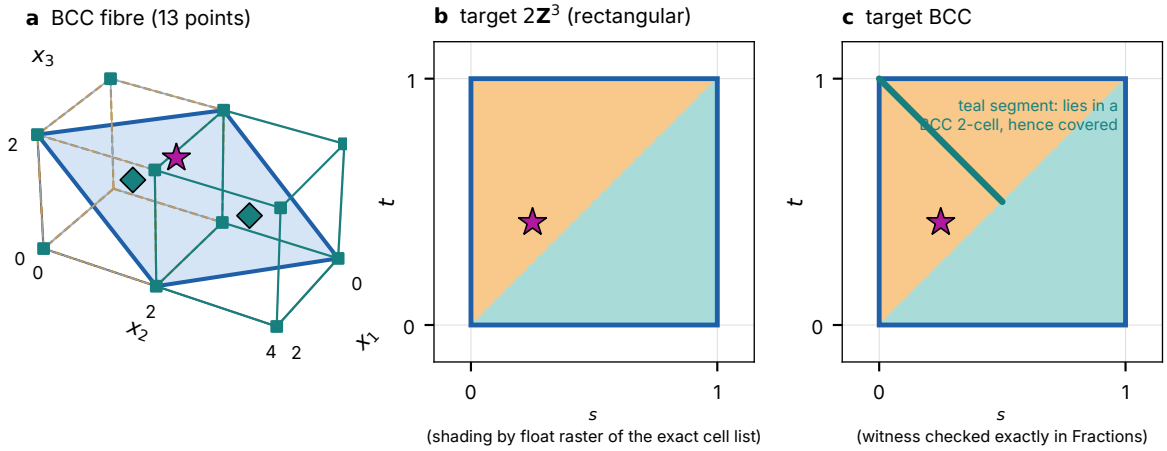


Figure 11: The body-centred target (Proposition 8.5; FINITE-EXACT for $Q = (2, 4, 2)$ only). The cell c of Figure 9 and its chart (s, t). (a) The 13 points of $F(\mathrm{BCC}, Q)$, $\mathrm{BCC} = \{z_1 \equiv z_2 \equiv z_3\ (2)\}$. (b) With the rectangular target $2\mathbb{Z}^3$ (f-vector $(11, 17, 8, 1)$) the uncovered region is the orange triangle T_0 of Figure 9. (c) With the BCC target (f-vector $(13, 33, 17, 1)$) the uncovered region is T_0 minus a teal segment, which lies in a BCC 2-cell and is therefore covered; but the exact witness $p = (\frac{5}{6}, \frac{5}{3}, \frac{3}{2})$ (chart coordinates $(\frac{1}{4}, \frac{5}{12})$, star) lies in no cell of $K_{\mathrm{abs}}(\mathrm{BCC}, Q)$. Shading is a floating-point raster (uncovered fractions ≈ 0.497 and ≈ 0.493 of the chart); the verdict at p is exact. This is a finite-exact example, not a classification of non-split targets.

Source record: `census` (`CW-0512`); status FINITE-EXACT; treatment FINITE.
These numbers are samples, not a classification theorem, and the release states explicitly that its earlier remark "no zero-vertex decomposition occurred" was an artefact of box scope. They are included only to indicate that preservation for non-split targets is genuinely dependent on the target lattice; the question of which non-split targets preserve U is open (Section 11).

9 Slabs: split targets and the corrected criterion

Suppose the target splits along the third coordinate with period k: $\Lambda' = \Lambda'_2 \oplus k\mathbb{Z}e_3$, with Λ'_2 a rank-two lattice in the (x_1, x_2)-plane. (The symbol k is the period here, not the dimension.) For $k = 1$ the coordinate is *free* ($e_3 \in \Lambda'$). Write $Q = (Q', q_3)$, $J = \lfloor q_3/k \rfloor$, $U = U(\Lambda'_2, Q')$ and $U^0 = U^0(\Lambda'_2, Q')$ (planar, origin excluded, resp. allowed). The three-dimensional union is a stack of planar unions with a single exceptional layer.

9.1 The union of a split target

Theorem 9.1 (slab structure)**.** *For $J \geq 1$, $U(\Lambda', Q) = \big[U \times [0, k]\big] \cup \big[U^0 \times [k, kJ]\big]$; for $J = 0$, $U(\Lambda', Q) = U \times \{0\}$.*

Source record: `P5-S4` (`CW-0511`); status PROVED; treatment FULL.

Proof. A nonzero vector (a,t) of Λ' with $a \neq 0 \neq t$ has $(a,0) \in \Lambda'$ conformally below it, so it is not Graver; $(a,0)$ is Graver in Λ' iff a is Graver in Λ'_2; $(0,t)$ is Graver iff $|t| = k$. So $\mathrm{Gr}(\Lambda') = \{(g,0) : g \in \mathrm{Gr}(\Lambda'_2)\} \cup \{\pm k e_3\}$ and every cell of Λ' is $c_2 \times \{z\}$ or $c_2 \times [z, z+k]$ with c_2 a planar cell (generators in $\mathrm{Gr}(\Lambda'_2)$), $z \in k\mathbb{Z}$, $z \geq 0$ (and $z + k \leq q_3$ in the second case). The corner (v,z) is zero iff $v = 0$ and $z = 0$. For $z = 0$ the planar factor must avoid the corner 0: c_2 is a cell of U; for $z \geq k$ the origin is allowed: c_2 is a cell of U^0. In the second form the corners with third coordinate $z + k \geq k$ impose nothing more. Taking the union over z: heights in $[0,k]$ are covered by cells with $z = 0$ (giving U at heights $< k$) and heights in $[k, kJ]$ by cells with $z \geq k$ (giving U^0); at height exactly k both occur and $U \subseteq U^0$. For $J = 0$ only $z = 0$ and no $\pm k e_3$ exist. □

Corollary 9.2 (free coordinate: slab lemma)**.** *If $e_3 \in \Lambda'$ ($k = 1$) and $q_3 \geq 1$ then for $0 < c < 1$: $(x', c) \in U(\Lambda', Q) \iff x' \in U(\Lambda'_2, Q')$; and for $c \in [1, q_3]$ the slice is U^0. If a point $p \in U(\Lambda, Q)$ has $0 < p_3 < 1$ and its planar part lies outside $U(\Lambda'_2, Q')$, then $U(\Lambda, Q) \not\subseteq U(\Lambda', Q)$ (slab obstruction).*

Source record: `P4-S1` (`CW-0492`); status PROVED; treatment FULL.

Source record: `P4-S2` (`CW-0494`); status PROVED; treatment ADAPTED.

Proof. Theorem 9.1 with $k = 1$. □

The slab lemma shows that a free-coordinate obstruction is inherited from the planar target: a planar gap times a thin slab. It is not a new three-dimensional coupling.

9.2 The corrected containment criterion

Theorem 9.3 (containment criterion, corrected form)**.** *Let $\Lambda \subseteq \Lambda' = \Lambda'_2 \oplus k\mathbb{Z}e_3$ and let c be a cell of $U(\Lambda, Q)$. Then*

$$c \subseteq U(\Lambda', Q) \iff \Big[\min_{x \in c} x_3 \geq k \Big] \text{ or } \Big[\mathrm{proj}_{12}\big(c \cap \{x_3 \leq k\}\big) \subseteq U(\Lambda'_2, Q') \Big].$$

Equivalently: $c \subseteq U(\Lambda', Q)$ iff $\mathrm{proj}_{12}(c \cap \{x_3 < k\}) \subseteq U$.

Source record: `P5-S4(ii)'` (`CW-0511`); status PROVED; treatment FULL.

Proof. The corners of c lie in $\Lambda \subseteq \Lambda'$, so their third coordinates lie in $k\mathbb{Z} \cap [0, q_3]$, and $c \subseteq \{0 \leq x_3 \leq kJ\}$. By Theorem 9.1 the slice of $U(\Lambda', Q)$ at height $z < k$ is U and at height $z \in [k, kJ]$ is U^0 (for $J = 0$ the cell lies in $\{x_3 = 0\}$ and the slice is U). Hence $c \subseteq U(\Lambda', Q)$ iff (a) $\mathrm{proj}_{12}(c \cap \{x_3 < k\}) \subseteq U$ and (b) $\mathrm{proj}_{12}(c \cap \{k \leq x_3 \leq kJ\}) \subseteq U^0$.

Claim: (b) always holds, indeed $\mathrm{proj}_{12}(c) \subseteq U^0$. The projection of $c = \Pi(u; g_1, \ldots, g_j)$ is the zonogon $\mathrm{proj}(u) + \sum [0,1]\mathrm{proj}(g_l)$ where $\mathrm{proj}(g_l) \in \Lambda'_2$ (the projection maps Λ' into Λ'_2); its corners are projections of corners of c, hence lie in $\Lambda'_2 \cap Q'$ (the origin allowed). Decompose each $\mathrm{proj}(g_l)$ by Lemma 7.1 into Graver vectors of Λ'_2 conformally below it and tile the zonogon by Lemma 7.2. As in the proof of Proposition 7.3 (Z.1), every tile vertex is the sum of the base and partial sums of the pieces; its coordinates lie between coordinates of corners of the projected cell, hence in $[0, Q']$, and it lies in Λ'_2. With the origin allowed, every tile is a cell of U^0, so $\mathrm{proj}(c) \subseteq U^0$.

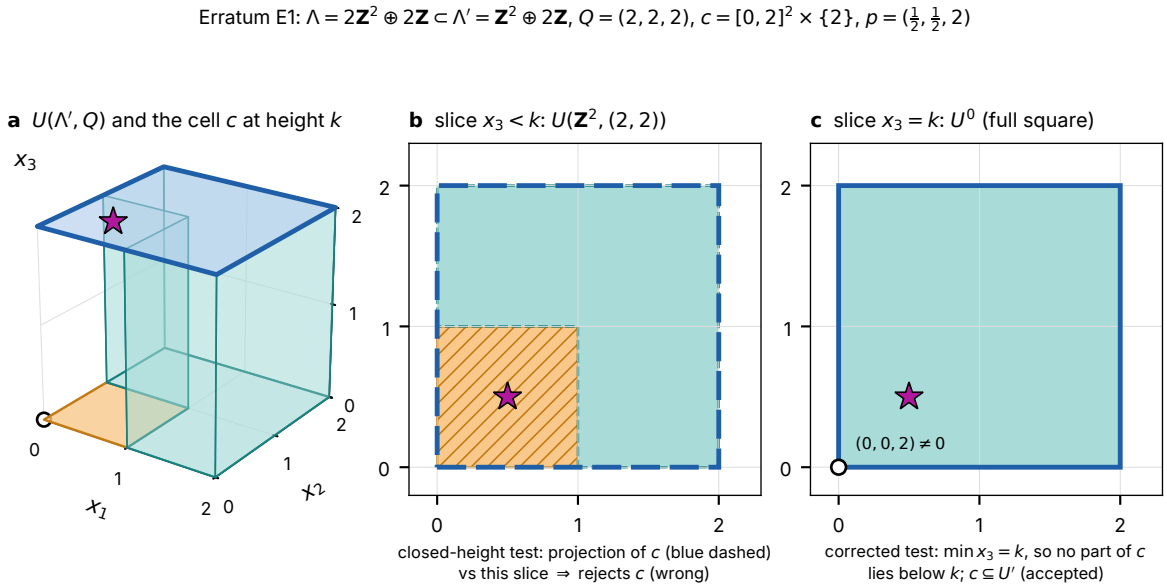


Figure 12: Why the strict-slab correction matters (Theorem 9.3, remark 9.4; $\Lambda = 2\mathbb{Z}^2 \oplus 2\mathbb{Z} \subset \Lambda' = \mathbb{Z}^2 \oplus 2\mathbb{Z}$, $k = 2$, $Q = (2,2,2)$; *exact computation*). The cell $c = [0,2]^2 \times \{2\}$ lies at height $x_3 = k$. (a) $U(\Lambda', Q)$ and c. (b) The slice $x_3 < k$ is the origin-excluded $U(\mathbb{Z}^2, (2,2))$, which does not contain $(\frac{1}{2}, \frac{1}{2})$. (c) The slice $x_3 = k$ is the origin-allowed full square U^0. The point $p = (\frac{1}{2}, \frac{1}{2}, 2)$ lies in $U(\Lambda', Q)$ (and all 121 grid points of c do), so $c \subseteq U(\Lambda', Q)$ and the corrected criterion (strict height $x_3 < k$; here $c \cap \{x_3 < k\} = \emptyset$) accepts c. The closed-height form of the first release would require $\mathrm{proj}_{12}(c) \subseteq U(\mathbb{Z}^2, (2,2))$, which fails at $(\frac{1}{2}, \frac{1}{2})$, and would wrongly reject c.

Therefore $c \subseteq U(\Lambda', Q)$ iff (a). If $c \cap \{x_3 < k\} = \emptyset$ (that is $\min x_3 \geq k$) (a) is vacuous. Otherwise $c \cap \{x_3 < k\}$ is a nonempty convex subset of the convex set c whose closure is $c \cap \{x_3 \leq k\}$ (move from any point of $c \cap \{x_3 \leq k\}$ towards a point with $x_3 < k$); since U is closed and the projection is continuous, (a) is equivalent to $\mathrm{proj}(c \cap \{x_3 \leq k\}) \subseteq U$. □

Remark 9.4 (the erratum). The first release of this criterion (v005, P5-S4(ii)) stated the condition $\mathrm{proj}_{12}(c \cap \{x_3 \leq k\}) \subseteq U$ *without* the alternative $\min x_3 \geq k$. That closed-height form is false, and it is *not* stated as valid here. Counterexample (erratum E1, ER-22, reproduced independently here): $\Lambda = 2\mathbb{Z}^2 \oplus 2\mathbb{Z}$, $\Lambda' = \mathbb{Z}^2 \oplus 2\mathbb{Z}$ (period $k = 2$), $Q = (2,2,2)$, $c = [0,2]^2 \times \{2\} = \Pi((0,0,2); (2,0,0), (0,2,0))$. Then c is a source cell and $c \subseteq U(\Lambda', Q)$ (for instance $(\frac{1}{2}, \frac{1}{2}, 2) \in U(\Lambda', Q)$ because at height $2 = k$ the target already contains the vertex $(0,0,2) \neq 0$, so its slice there is U^0, not U); but the closed-height criterion would reject c because $(\frac{1}{2}, \frac{1}{2}) \notin U(\mathbb{Z}^2, (2,2))$. All earlier consequences of the *hard-cell* comparisons were unaffected: a hard cell has $\min x_3 = 0 < k$, where the closed form is correct.

9.3 The free-coordinate case

Corollary 9.5 (complete slab criterion)**.** *Let $e_3 \in \Lambda'$ ($k = 1$), $q_3 \geq 1$ and $\Lambda \subseteq \Lambda'$. Then $U(\Lambda, Q) \subseteq U(\Lambda', Q)$ iff there is no point $p \in U(\Lambda, Q)$ with $p_3 < 1$ whose planar part lies outside $U(\Lambda'_2, Q')$.*

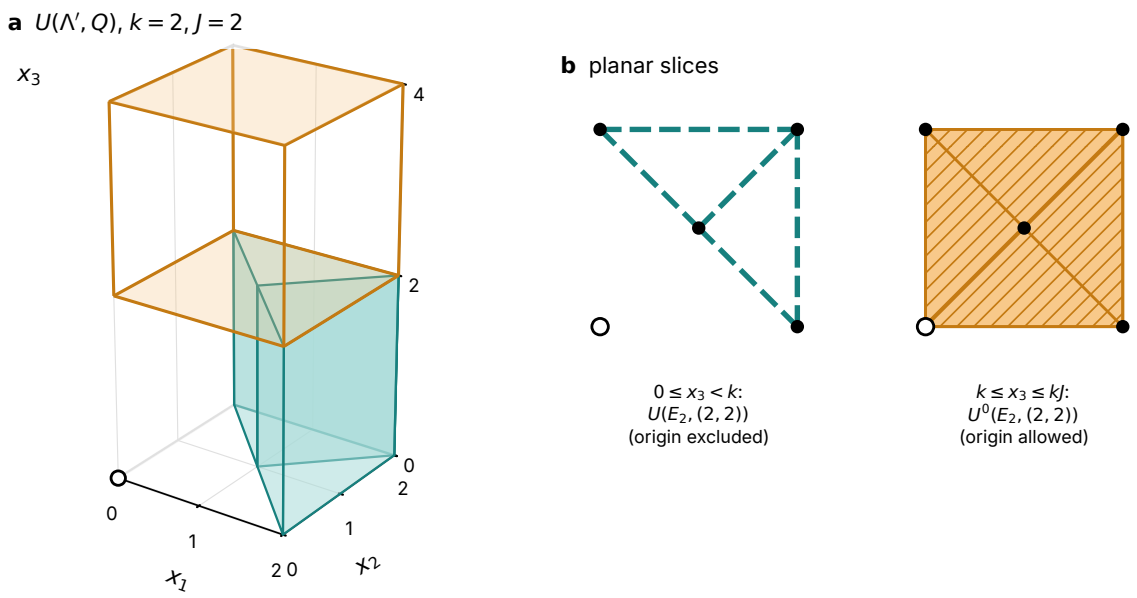


Figure 13: Geometry of a split target (Theorem 9.1; $\Lambda' = E_2 \oplus 2\mathbb{Z}e_3$, $Q = (2,2,4)$, $k = 2$, $J = 2$; planar complexes *exact*, vertical stacking *schematic*). (a) The union $U(\Lambda', Q)$: below height k the slice is the origin-excluded planar union $U = U(E_2, (2,2))$ (blue; one-dimensional here, 5 edges and no square), from height k upward it is the origin-allowed union U^0 (orange; the filled square $[0,2]^2$ and 8 edges). (b) The two planar slices. A source cell is contained iff its part below height k projects into U. The figure shows a realised union, not an abstract complex.

Source record: `P4-S3` (`CW-0498`); status PROVED; treatment ADAPTED.

Proof. Theorem 9.3 with $k = 1$, applied to every cell: $c \subseteq U'$ iff $\mathrm{proj}(c \cap \{x_3 < 1\}) \subseteq U$. □

In particular, for free-coordinate targets every failure is a slab obstruction (the "S1 failures" of the intermediate releases do not exist), and the failures of this class are exactly the planar target gaps seen through the layer $0 < x_3 < 1$. The slab criterion is drawn in Figure 13.

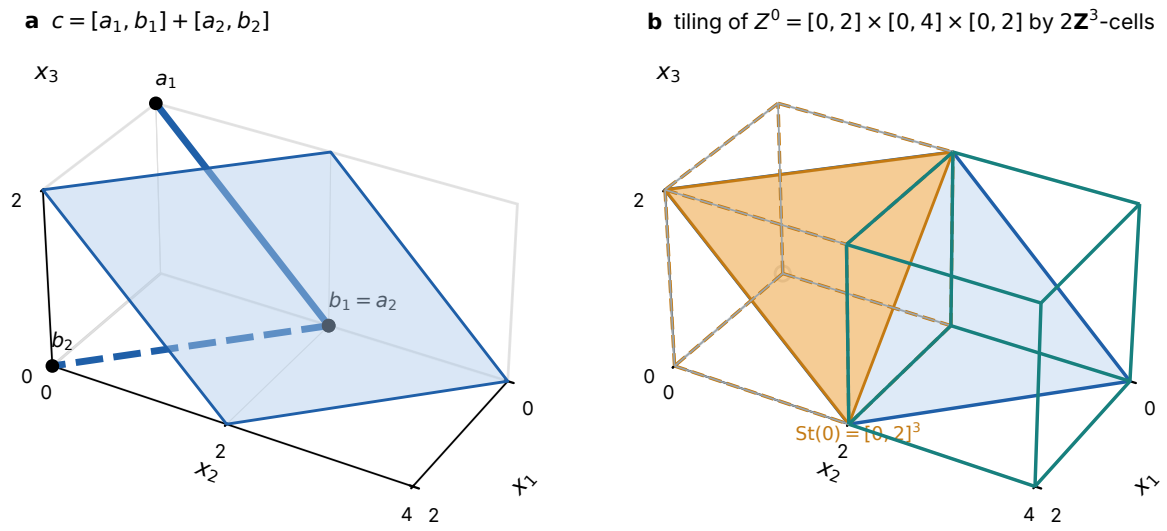
10 Hard cells and a coupled-target specimen

10.1 Structure of a hard cell

Proposition 10.1 (a hard cell is a Minkowski sum of segments)**.** *Let $\Lambda \subseteq \Lambda'$ and let $c = \Pi(u; g_1, \dots, g_j)$ be a hard cell of $U(\Lambda, Q)$ in the sense of Proposition 7.3. Put $a_i = -g_i^- \geq 0$ and $b_i = g_i^+ \geq 0$; both lie in Λ', have disjoint supports and $g_i = b_i - a_i$. Then $u = \sum a_i$ and*

$$c = \sum_i [a_i, b_i] \;\subseteq\; Z^0 := \sum_i \big([0, a_i] + [0, b_i]\big) \;\subseteq\; [0, Q_c],$$

$$Q_c = \sum_i (a_i + b_i) \ \textit{(the top corner of } c\textit{)}.$$



Every tile of the tiling not containing 0 is a cell of $U(\Lambda', Q_c)$; only the triangle $T_0 = c \cap \mathrm{St}(0)$ (orange) can lie outside it.

Figure 14: Minkowski decomposition of a hard cell and the origin star (Proposition 10.1; the cell c of Theorem 8.2, $\Lambda = 2E_3$, $\Lambda' = 2\mathbb{Z}^3$; *exact computation*). (a) $c = [a_1, b_1] + [a_2, b_2]$ with $a_1 = (0, 0, 2)$, $b_1 = (0, 2, 0)$, $a_2 = (0, 2, 0)$, $b_2 = (2, 0, 0)$ and $u = a_1 + a_2 = (0, 2, 2)$, drawn as the sum of two segments. (b) The zonotope $Z^0 = [0, 2] \times [0, 4] \times [0, 2]$ tiled by $2\mathbb{Z}^3$-cells: the tile $[0, 2]^3$ has the origin as a vertex and is the star $\mathrm{St}(0)$ (orange; it is not a cell of $K_{\mathrm{abs}}(2\mathbb{Z}^3, Q)$ because the origin is not in the fibre), the tile $[0, 2] \times [2, 4] \times [0, 2]$ is a cell of the target. The only part of c not covered by non-star tiles lies in $\mathrm{St}(0)$; this is the region T_0 of Figure 9.

In a fine tiling of Z^0 by Graver pieces of the a_i, b_i (Lemmas 7.1 and 7.2) every tile that does not contain the origin as a vertex is a cell of $U(\Lambda', Q_c)$; the origin is the only vertex that is not a nonzero subset sum. Hence the only part of c that can fail to lie in $U(\Lambda', Q_c)$ is $c \cap \mathrm{St}(0)$, the star of the origin in the tiling.

Source record: `E2P4v5-H1` (`CW-0503`); status PROVED; treatment FULL.

Proof. $m(c) = u + \sum g_i^- = 0$ gives $u = \sum a_i$, and $u + \sum \lambda_i g_i = \sum_i [(1 - \lambda_i)a_i + \lambda_i b_i]$, which is a point of $\sum [a_i, b_i] \subseteq Z^0$. The pieces of the a_i, b_i are nonnegative nonzero vectors, so a subset sum vanishes only for the empty subset; the vertices of tiles of Z^0 are subset sums and the remaining argument is that of Proposition 7.3(Z.1) with s_i the partial sums. Coordinates stay in $[0, Q_c]$ because all pieces are ≥ 0. □

This is the precise form of the statement "the deleted origin matters only for cells that touch every coordinate plane". In the plane the star of the origin is covered by the coarser rectangular tiling (the proof of Theorem 7.5); in the examples of Section 8 the star of the origin is the corner cube.

10.2 The coupled specimen P4-F110

The previous sections concern general lattices. We record the one explicit coupled collective transition for which the whole transport can be decided for every box size m.

Proposition 10.2 (arithmetic of P4-F110)**.** *For every integer $m \geq 1$, $x_m = 2 \cdot 110^{4m}/9$ has $L = 110^{2m}$, $G = 9$, $\mathcal{C}(x_m) = 110^{4m}/3 =: y_m$; the active primes $(2, 5, 11)$ and the box $Q_m = [0, 2m]^3$ are the same for x_m and y_m. The relation lattices are $\Lambda_9 = \{(a, b, c) : a - b + c \equiv 0\ (6)\}$ and $\Lambda_3 = E_3$, with $\Lambda_9 \subsetneq \Lambda_3$; the orders modulo 9 are $(6, 6, 6)$ and modulo 3 are $(2, 2, 2)$, and both lattices are non-rectangular.*

Source record: `P4-A` (`CW-0483`); status PROVED; treatment FULL.

Source record: `P4-L` (`CW-0484`); status PROVED; treatment FULL.

Proof. $\mathrm{num}(x_m) = 2^{4m+1}5^{4m}11^{4m}$, so $d^2 \mid \mathrm{num}$ iff $d = 2^a5^b11^c$ with $a, b, c \leq 2m$. Modulo 9: $5 \equiv 2^{-1}$, $11 \equiv 2$, $\mathrm{ord}_9(2) = 6$, so $d \equiv 2^{a-b+c}$ and $d \equiv 1$ iff $a - b + c \equiv 0\ (6)$. Every $d \in D$ has $9 \mid d - 1$, so $9 \mid G$. The vectors $(1, 1, 0)$ and $(0, 1, 1)$ give $10, 55 \in D$ with $\gcd(9, 54) = 9$, so $G = 9$. The vectors $(2m, 2m, 0)$, $(0, 2m, 2m)$ give $10^{2m}, 55^{2m} \in D$, whose lcm is 110^{2m}, and every $d \in D$ divides 110^{2m}; so $L = 110^{2m}$. Then $\mathcal{C}(x_m) = (2 + 1) \cdot 110^{4m}/9 = 110^{4m}/3$; the numerator 110^{4m} has $v_p = 4m$ for $p = 2, 5, 11$, so the primes and the box $[0, 2m]^3$ are unchanged. Modulo 3 all of $2, 5, 11$ are -1, so $\chi_3 = (-1)^{a+b+c}$, and $a - b + c \equiv 0\ (6)$ implies $a + b + c = (a - b + c) + 2b$ even; $(2, 0, 0) \in E_3 \setminus \Lambda_9$. □

Theorem 10.3 (hull theorem for E_3)**.** *Let $n \geq 4$ be even and $Q = [0, n]^3$. Then $U(E_3, Q) = \mathrm{conv}\, F(E_3, Q) = \{z \in [0, n]^3 : z_1 + z_2 + z_3 \geq 2\}$. For $n = 2$ this fails.*

Source record: `P4-T1` (`CW-0485`); status PROVED; treatment ADAPTED.

Proof. (1) *Hull.* F contains all cube vertices except 0 (their coordinate sums are even) and $(2, 0, 0), (0, 2, 0), (0, 0, 2)$, and every point of F has coordinate sum ≥ 2. The polytope $[0, n]^3 \cap \{\sum \geq 2\}$ has exactly these 10 vertices, so $\mathrm{conv}\, F$ equals it. (2) *Cube tiling.* $2e_i \in \mathrm{Gr}(E_3)$ (Proposition 4.4). For $w \in (2\mathbb{Z})^3 \cap [0, n - 2]^3$, $w \neq 0$, the eight corners $w + \{0, 2\}^3$ have even sum, are nonzero and lie in the box, so the cube is a 3-cell; these cubes tile $[0, n]^3$ minus the corner cube $[0, 2]^3$. (3) *The corner cube.* We must show $[0, 2]^3 \cap \{\sum \geq 2\} \subseteq U(E_3, [0, n]^3)$. The Graver set of E_3 in $[-n, n]^3$ is the same for every $n \geq 2$ (a conformally minimal vector lies in the box of its own size) and $F(E_3, 4) \subseteq F(E_3, n)$, so the cells of the box-4 complex are cells of the box-n complex, and it suffices to take $n = 4$. For $n = 4$ the argument is a finite exact certificate, identified precisely in [6]: the object `n4_target_hull_cover` in `results/P4_CERTIFICATES.json` of the release `GEO_HD_E2_P4_v001` (byte-identical in `v001`–`v004`), replayed by shard `S4` of that release. It is a decomposition of $\mathrm{conv}\, F(E_3, [0, 4]^3)$ into pieces each lying in a single maximal cell of $K_{\mathrm{abs}}(E_3, [0, 4]^3)$, with no gap nodes; it is a statement about one explicit polytope and one explicit finite set of cells. *Logical role.* The certificate is used for exactly one implication: $[0, 2]^3 \cap \{\sum x_i \geq 2\} \subseteq U(E_3, [0, 4]^3)$, from which part (3) follows for all even $n \geq 4$ by the Graver-set argument above. Parts (1) and (2) do not use it. All of Theorem 10.3, and everything that cites it (Theorem 10.4 and Corollary 10.5), is therefore PROVED *modulo this finite exact computation*, and this status is recorded in Section 11. *Independent replay.* Because the certificate is cited and not reproduced, the present manuscript supplies a separate exact check of the same implication (`scripts/replay_n4_cover.py`, output

`results/N4_COVER_REPLAY.json`): in exact rational arithmetic it enumerates the 764 three-dimensional cells of $K_{\mathrm{abs}}(E_3, [0,4]^3)$ and subtracts them from the polytope $[0,2]^3 \cap \{\sum x_i \geq 2\}$ by exact polytope subtraction; no piece remains. The same code applied to the box $[0,2]^3$ leaves 9 pieces, the gap at $n = 2$ (the release reports 94 gap pieces for its own decomposition; piece counts depend on the decomposition used). This replay uses the cells of this manuscript's enumeration and is a manuscript-level check of the implication, not an independent verification of the release. As a further sanity check only, all 609 points of the quarter-grid of $[0,2]^3$ with coordinate sum ≥ 2 lie in $U(E_3, [0,4]^3)$. For $n = 2$ the statement fails: the point $(0, 1, \frac{3}{2})$ lies in $\operatorname{conv} F$ (its sum is $\frac{5}{2} \geq 2$) but in no cell of $K_{\mathrm{abs}}(E_3, [0,2]^3)$ (exact enumeration here; the release certifies the gap at $n = 2$ with 94 gap pieces). This is consistent with the planar face $x_1 = 0$, where E_3 restricts to E_2 in the exceptional hollow box $(2, 2)$. □

Theorem 10.4 (transport for P4-F110)**.** *For every $m \geq 1$: $U(\Lambda_9, Q_m) \subsetneq U(E_3, Q_m)$, the transport class is "source contained", and $U(E_3, Q_m) \not\subseteq U(\Lambda_9, Q_m)$.*

Source record: `P4-T2` (`CW-0486`); status PROVED-SCOPED; treatment ADAPTED.

Proof. For $m \geq 2$ ($n = 2m \geq 4$): $U(\Lambda_9) \subseteq \operatorname{conv} F(\Lambda_9) \subseteq \operatorname{conv} F(E_3) = U(E_3)$ by $F(\Lambda_9) \subseteq F(E_3)$ and Theorem 10.3. Strictness: $(2,0,0)$ is a vertex of $\operatorname{conv} F(E_3)$ lying in $F(E_3)$ but not in $F(\Lambda_9)$ ($2 - 0 + 0 = 2 \not\equiv 0$ (6)); a vertex of $\operatorname{conv} F(E_3)$ that lies in $\operatorname{conv} F(\Lambda_9) \subseteq \operatorname{conv} F(E_3)$ is a vertex of $\operatorname{conv} F(\Lambda_9)$, hence in $F(\Lambda_9)$; so $(2,0,0) \notin U(\Lambda_9)$ (indeed $U(\Lambda_9) \subseteq \operatorname{conv} F(\Lambda_9)$). For $m = 1$ ($n = 2$, where Theorem 10.3 fails) the release gives direct exact certificates (the two maximal source cells covered by 18 target maximal cells, independently replayed); this finite certificate is cited. The reverse non-containment is the same vertex argument. □

The scope of the theorem is the family P4-F110 only. A direct engine verification beyond $m = 2$ was not attempted by the release (cost); the case $m \geq 3$ is covered by Theorem 10.3.

Corollary 10.5 (predictive criterion for E_3 targets)**.** *Let $\Lambda \subseteq E_3$ be any sublattice and $n \geq 4$ even. Then $U(\Lambda, [0,n]^3) \subseteq U(E_3, [0,n]^3)$.*

Source record: `P4-C1` (`CW-0489`); status PROVED; treatment FULL.

Proof. $U(\Lambda) \subseteq \operatorname{conv} F(\Lambda) \subseteq \operatorname{conv} F(E_3) = U(E_3)$ by Theorem 10.3. □

The first release of this result added an equality clause (equality of unions iff $\operatorname{conv} F(\Lambda)$ contains the ten vertices of Theorem 10.3); that clause is withdrawn (counterexample $\Lambda = 2\mathbb{Z}^3$, $n = 4$) and only the containment is stated (ER-21).

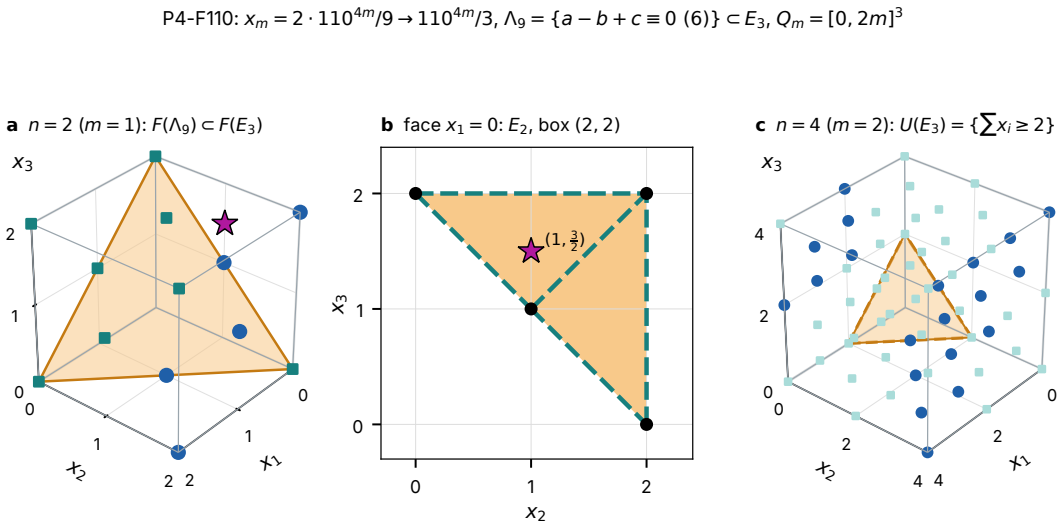
Source record: `P4-C1 (v001)` (`CW-0487`); status CORRECTED; treatment EXCLUDED.

Its predictive cost is $O(|F|)$ lattice-point tests with no cell reconstruction.

Proposition 10.6 (holes are not obstructions; FINITE-EXACT, one specimen)**.** *For $m = 1$ the target E_3 is not hull-covered (Theorem 10.3, $n = 2$) and yet $U(\Lambda_9, Q_1) \subseteq U(E_3, Q_1)$.*

Source record: `P4-O1` (`CW-0488`); status FINITE-EXACT; treatment FINITE.

This is an observation about one specimen (the release certifies it exactly with 94 gap pieces); it shows only that gaps of the target do not by themselves obstruct containment.



(a) blue: $F(\Lambda_9)$, teal squares: $F(E_3) \setminus F(\Lambda_9)$; the orange triangle $x_1 + x_2 + x_3 = 2$ is the cut of conv $F(E_3)$, and the star is the exact gap point $(0, 1, \frac{3}{2})$ lying in no cell. (b) its trace in the planar hollow-box face. (c) for $n = 4$ the union equals conv $F(E_3)$ (hull theorem; $n = 4$ by the cited certificate), so every source union is contained.

Figure 15: The coupled family P4-F110 ($x_m = 2 \cdot 110^{4m}/9 \to 110^{4m}/3$, $\Lambda_9 = \{a - b + c \equiv 0\ (6)\} \subset E_3$, $Q_m = [0, 2m]^3$; *exact computation* for the data, theorem-level for the regime in (c)). (a) $n = 2$ ($m = 1$): $F(\Lambda_9, Q_1)$ (5 points) inside $F(E_3, Q_1)$ (13 points); E_3 is not hull-covered (Theorem 10.3), the point $(0, 1, \frac{3}{2})$ is in conv $F(E_3)$ but in no cell of $K_{\text{abs}}(E_3, Q_1)$. (b) The face $x_1 = 0$, where E_3 restricts to E_2 in the hollow box $(2, 2)$ and the gap point appears as $(1, \frac{3}{2})$. (c) $n = 4$ ($m = 2$): $U(E_3, Q_2) = \{\sum x_i \geq 2\}$ by Theorem 10.3, with $|F(E_3)| = 62$ and $|F(\Lambda_9)| = 20$. For $m = 1$ containment $U(\Lambda_9) \subseteq U(E_3)$ nevertheless holds (Proposition 10.6); for $m \geq 2$ it follows from Theorem 10.3 (see Theorem 10.4), whose $n = 4$ step rests on the certificate of Theorem 10.3(3). No claim is drawn from the picture alone.

11 Status, limits, and open problems

11.1 What is proved and what is not

result	status	scope
lattice transport (Theorem 3.2, Proposition 3.1)	PROVED	all lattices Λ_b
abstract cell transport (Theorem 4.3)	PROVED	all lattices, common box
three-stage orbit (Theorem 4.5, Corollary 4.6)	PROVED	all $k \geq 2$, primes $\equiv 2$ (3), $2 \in \mathcal{P}$
planar family (Theorem 4.8)	PROVED	$x_h = 21^{2h}/5$, $h \geq 1$
corner cube (Theorem 5.2), even-sum (Theorem 5.4)	PROVED	rectangular targets, any k; even-sum $k = 3$
planar monotonicity (Theorem 7.5)	PROVED	$k = 2$, nested lattices, every box (Lemma 7.2 proved in appendix B)
dilation family, no free coordinate (Theorem 8.2)	PROVED	$k = 3$
slab structure and criterion (Theorems 9.1 and 9.3)	PROVED	split targets, period k
hull theorem E_3 (Theorem 10.3)	PROVED mod. cert.	$n \geq 4$ even; $n = 4$ certificate [6] cited, replayed in Section 10
BCC failure (Proposition 8.5)	FINITE-EXACT	$Q = (2, 4, 2)$
census (Proposition 8.7)	FINITE-EXACT	stated sample, not a theorem
specimen counts and Betti numbers (Propositions 4.7 and 6.3)	FINITE-EXACT	the named specimens

Inherited and cited without reproof: the fibre bijection input (Foundations Vol. I Thm. 1.2), $D(x) \subseteq D(\mathcal{C}(x))$ (F.T31), the denominator facts behind E2-3, the planar normal form and G12-T8 (GEO-I) and the finite $n = 4$ certificate of Theorem 10.3. The lifting fact formerly cited as SF is proved in appendix B. Inherited verification of the frozen releases was not re-run; the exact checks of this manuscript are independent re-computations of the worked examples (`scripts/run_exact_checks.py`) and do not constitute external verification.

11.2 Open problems

Problem 11.1 (which non-split targets preserve U)**.** Classify the lattices $\Lambda' \subseteq \mathbb{Z}^3$ with no free and no split coordinate for which $U(\Lambda, Q) \subseteq U(\Lambda', Q)$ for *every* $\Lambda \subseteq \Lambda'$ and every box. The census (Proposition 8.7) shows that 77 of 84 sampled targets of index 2–6 show no failure while BCC and six index-6 lattices fail; no property of the target that separates the two groups is known (release item P5-Q1).

Problem 11.2 (collective transitions with incomparable lattices)**.** For an admissible planar transition with $F \subseteq F'$ but $\Lambda \not\subseteq \Lambda'$ (which occurs, Proposition 3.6), decide whether $U \subseteq U'$. Theorem 7.5 does not apply, and Lemma 5.1 gives containment only in the G12-T8 regime.

Problem 11.3 (dimension $k \geq 4$ and exact classification)**.** Determine the failing cells and the minimal failing boxes for $k \geq 4$ and for non-even-sum sources; Theorem 5.2(c) bounds the shape of a scaled vertex but does not decide which boxes fail.

Problem 11.4 (embeddedness)**.** Characterise when $K_{\mathrm{abs}}(\Lambda, Q)$ is embedded (so that $K_{\mathrm{emb}} = K_{\mathrm{abs}}$ and topological statements about U and K_{abs} coincide). For the specimen E_3, $Q = [0,2]^3$ it is not embedded and $b_2(U) = 18 \neq b_2(K_{\mathrm{abs}}) = 6$.

Problem 11.5 (tiling lemma; resolved in v0.2)**.** Remove the dependence of Lemma 7.2 on the external standard fact SF by a self-contained proof of the covering property of the lower facets. *Resolved:* the proof is given in appendix B. The item is kept so that the numbering of the first version is unchanged.

Problem 11.6 (parameter-family classification and coupled-target theory)**.** The releases name two programmes that were never started: a classification of transport for parameter families beyond P4-F110 (P4-T3), and an essentially three-dimensional theory of coupled-target obstruction (P4-T4). Neither is addressed here. The prior-art search accompanying the releases was limited and is not a novelty clearance; no priority claim is made for any statement of this manuscript.

The open items recorded at v002 and v003 (`CW-0493`, `CW-0495`) comprise: planar containment (resolved by Theorem 7.5); the converse of the slab obstruction and the non-existence of "S1 failures" (resolved by Corollary 9.5); preservation for targets without a free coordinate (resolved negatively by Theorem 8.2); and P4-T3, P4-T4 and the prior-art audit (problem 11.6).

Source record: `OPEN(v002)` (`CW-0493`); status OPEN; treatment OPEN.

Source record: `OPEN(v003)` (`CW-0495`); status OPEN; treatment OPEN.

11.3 Conclusion

Transport of occurrence geometry has three distinct levels. Vertices are transported by the lattice inclusion alone. Cells of the abstract complex are transported exactly when their vertices lie in the target fibre and their generators remain Graver; this can fail for every cell of positive dimension at once. The realised union is transported in the plane for every lattice inclusion and every box, because a cell that touches both axes has its geometry forced by the arithmetic of the axis orders; in dimension three the same inclusion can fail in infinite families, the failure is carried by cells touching all three coordinate planes, and for targets that split along one coordinate the failure is decided exactly by a planar condition on the part of the cell below height k. What remains unknown is the full classification in dimension three for non-split targets and the behaviour beyond dimension three.

References

[1] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume I: Generative Arithmetic and Structural Reachability* (Theorem 1.2, the occurrence set). Source: `ukbbi/UNNS`, commit `9a021346b2b3d964523440d2737b9d0c4ddbb3e5`, `projects/UNNS_MATH_FOUNDATIONS/Manuscript`. https://github.com/ukbbi/UNNS/tree/9a021346b2b3d964523440d2737b9d0c4ddbb3e5/projects/UNNS_MATH_FOUNDATIONS/Manuscript

[2] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume II: Collective Dynamics, Prime Rigidity, and Effective Predictability* (Theorem 1.3 = F.T31, $D(x) \subseteq D(\mathcal{C}(x))$). Same commit and path as [1].

[3] Foundations reference collection v001 (SELECT-I/II, FIELD-I v001–v012; frozen), `ukbbi/UNNS`, commit `bb8b12d753c600c597feb00df17fe139c7cc65fa` (23 Oct. 2026), `projects/UNNS_MATH_FOUNDATIONS/sources`. https://github.com/ukbbi/UNNS/tree/bb8b12d753c600c597feb00df17fe139c7cc65fa/projects/UNNS_MATH_FOUNDATIONS/sources

[4] GEO-I releases v001–v012 (planar normal form: v004 S1; planar stabilisation theorem G12-T8: v012). `ukbbi/UNNS`, commit `9a021346b2b3d964523440d2737b9d0c4ddbb3e5`, tree `projects/UNNS_Higher-Dimensional Geometry`. https://github.com/ukbbi/UNNS/tree/9a021346b2b3d964523440d2737b9d0c4ddbb3e5/projects/UNNS_Higher-Dimensional%20Geometry

[5] GEO-HD-1 research collection, snapshot `a88f31487f3b05c9db297dcd1192d56eea2b0c7d` (the pin named in the programme specification; not independently verified in this work). The source releases of the records of this manuscript, with the SHA-256 of the release archives as recorded in the programme:

- `GEO_HD_E2_P1_v001.zip` `2463492df1c829c5013ffa2838c9b710f7db20fe172bad80d62c0dd2798f819b` (E2-1..E2-11);
- `GEO_HD_E2_P2_v001.zip` `5680ab6c0b1c27bd56296ef7ada2924c6db9da695221dc7a4e1ebe2c1c9cd847` (P2-series);
- `GEO_HD_E2_P3_v001.zip` `7709f4f7bcf206c204634fb5b88ad69e70bef55ed910f5b28ef50f25165dd0f9` (P3-series);
- `GEO_HD_E2_P4_v001.zip` `6e7190bc03c4f04c9ae032d323ab988816b1214ef3a71530d6ce450244fb7fa6` (P4-series; contains the certificate of Theorem 10.3, see [6]);
- `GEO_HD_E2_P4_v002.zip`, `_v003.zip`, `_v004.zip`, `_v005.zip` (later P4 releases; P5-series in `v005`) and `GEO_HD_E2_P4_v005_ERRATA_E1.zip` `f41a353b71e57bc6df3a7f5858642aea4ff71aa6078ecebbb3e2a6704de6f8b3` (erratum E1, remark 9.4).

[6] Exact finite certificate used in Theorem 10.3, $n = 4$: file `results/P4_CERTIFICATES.json`, key `n4_target_hull_cover` of the release `GEO_HD_E2_P4_v001` (the file is byte-identical in `v001`–`v004`; SHA-256 `5dd01aa86d878bdd53ba31bce71f2a6fba9aa33ce965a53c04b30e0de2682002`, 808 345 bytes), replayed by shard `S4` of the release's `RUN_VERIFY.py`. See Section 10 for its logical role and for the independent replay of this manuscript.

[7] J. E. Graver, On the foundations of linear and integer linear programming I, *Math. Programming* **9** (1975), 207–226.

[8] B. Sturmfels, *Gröbner Bases and Convex Polytopes*, University Lecture Series 8, American Mathematical Society, 1996.

[9] J. A. De Loera, R. Hemmecke, M. Köppe, *Algebraic and Geometric Ideas in the Theory of Discrete Optimization*, MOS–SIAM Series on Optimization 14, SIAM, 2013.

[10] G. M. Ziegler, *Lectures on Polytopes*, Graduate Texts in Mathematics 152, Springer, 1995 (zonotopes, subdivisions and liftings).

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[12] J. Richter-Gebert, G. M. Ziegler, Zonotopal tilings and the Bohne–Dress theorem, in: *Jerusalem Combinatorics '93*, Contemporary Mathematics 178, American Mathematical Society, 1994, 211–232.

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[14] A. Björner, M. Las Vergnas, B. Sturmfels, N. White, G. M. Ziegler, *Oriented Matroids*, 2nd ed., Encyclopedia of Mathematics and its Applications 46, Cambridge University Press, 1999 (zonotopal tilings).

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On the references. UNNS sources are cited at the commits fixed by the programme, all repository access during this work was read-only, and the URLs above follow GitHub's standard pattern; they were not fetched from the environment in which the manuscript was prepared. The external literature is cited by author, title and publication data; no section numbers of books are quoted, because the arguments of this manuscript that use them are proved in the text (appendix B) or are standard facts of the theory of Graver bases and zonotopes.

A Theorem map to the crosswalk

Every source record allocated to this manuscript (57 rows of the HD-1B line) with its crosswalk row, original theorem identifier, status preserved from the source, proof treatment in this manuscript, and location. Treatments: FULL = full proof here; ADAPTED = proof adapted from the source with its inherited inputs cited; CITED = inherited result cited under its hypotheses; EXAMPLE = used as a worked example or counterexample; FINITE = finite-exact evidence only; OPEN = unresolved; EXCLUDED = not needed (reason in the allocation table).

CW row	source id	status	treatment	section	number
CW-0456	E2-1	PROVED	FULL	4	4.5
CW-0457	E2-2	PROVED	FULL	3	3.1
CW-0458	E2-3	PROVED	ADAPTED	3	3.5
CW-0459	E2-4	PROVED	ADAPTED	3	3.5
CW-0460	E2-5	PROVED	FULL	4	4.4
CW-0461	E2-6	PROVED	FULL	4	4.6
CW-0462	E2-7	PROVED	ADAPTED	5	5.5
CW-0463	E2-8	PROVED	FULL	6	6.1
CW-0464	E2-10	FINITE-EXACT	FINITE	4	4.7
CW-0464	E2-10	FINITE-EXACT	FINITE	6	6.2
CW-0465	E2-11	COUNTEREXAMPLE	EXAMPLE	6	6.4
CW-0466	P2-L1	PROVED	FULL	4	4.1
CW-0467	P2-T1	PROVED	FULL	3	3.2
CW-0467	P2-T1	PROVED	FULL	3	3.3
CW-0468	P2-L2	PROVED	FULL	3	3.4
CW-0469	P2-T2	PROVED	FULL	4	4.2
CW-0470	P2-T3	PROVED	FULL	4	4.3
CW-0471	P2-TF1	PROVED	FULL	4	4.8
CW-0472	P2-D	FINITE-EXACT	FINITE	3	3.6
CW-0473	CP2	OPEN	EXCLUDED	7	7.7
CW-0474	P2-G	FINITE-EXACT	FINITE	6	6.3
CW-0475	E2-P2-V	PROVED	EXAMPLE	6	6.4
CW-0476	P3-T0	PROVED	FULL	5	5.1
CW-0477	P3-L1	PROVED	CITED	7	7.7
CW-0478	P3-C1	PROVED	CITED	7	7.7
CW-0479	P3-T3	PROVED	FULL	5	5.2
CW-0480	P3-T3(h)	PROVED	FULL	5	5.4
CW-0481	P3-T4	PROVED	EXCLUDED	5	5.6
CW-0482	CP3-A	OPEN	EXCLUDED	7	7.7
CW-0483	P4-A	PROVED	FULL	10	10.2
CW-0484	P4-L	PROVED	FULL	10	10.2
CW-0485	P4-T1	PROVED	ADAPTED	10	10.3
CW-0486	P4-T2	PROVED-SCOPED	ADAPTED	10	10.4
CW-0487	P4-C1 (v001)	CORRECTED	EXCLUDED	10	10.5
CW-0488	P4-O1	FINITE-EXACT	FINITE	10	10.6
CW-0489	P4-C1	PROVED	FULL	10	10.5
CW-0490	P4-Z1	PROVED	FULL	7	7.3
CW-0491	P4-Z1.a	PROVED	FULL	7	7.3
CW-0492	P4-S1	PROVED	FULL	9	9.2
CW-0493	OPEN(v002)	OPEN	OPEN	11	11.6
CW-0494	P4-S2	PROVED	ADAPTED	9	9.2
CW-0495	OPEN(v003)	OPEN	OPEN	11	11.6
CW-0496	P4-Z2	PROVED	FULL	7	7.3
CW-0497	P4-P1	PROVED	FULL	7	7.5
CW-0498	P4-S3	PROVED	ADAPTED	9	9.5
CW-0499	E2P4v5-L1	PROVED	FULL	7	7.1
CW-0500	E2P4v5-T0	PROVED	ADAPTED	7	7.2
CW-0501	E2P4v5-Z	PROVED	FULL	7	7.3
CW-0502	E2P4v5-LG	PROVED	FULL	7	7.4
CW-0503	E2P4v5-H1	PROVED	FULL	10	10.1
CW-0504	P5-P1	PROVED	FULL	7	7.5
CW-0505	P5-P1.a	PROVED	FULL	7	7.8
CW-0506	P5-F1	PROVED	FULL	8	8.6
CW-0507	P5-D1	PROVED	FULL	8	8.1
CW-0508	P5-C1	PROVED	FULL	8	8.2
CW-0509	P5-C2	PROVED	FULL	8	8.3
CW-0510	P5-C3	FINITE-EXACT	FINITE	8	8.5
CW-0511	P5-S4	PROVED	FULL	9	9.1
CW-0511	P5-S4(ii)’	PROVED	FULL	9	9.3
CW-0512	census	FINITE-EXACT	FINITE	8	8.7

B Proof of the lifting lemma used in Lemma 7.2

The tiling lemma (Lemma 7.2) uses the standard statement that the lower facets of a lifted polytope project onto its shadow with disjoint interiors (the regular subdivision induced by a lifting, [10, 11]; the construction behind the Bohne–Dress correspondence [15, 12, 13, 14]). For completeness the argument is given here.

Let $\pi : \mathbb{R}^{d+1} \to \mathbb{R}^d$ forget the last coordinate and let $\tilde{P} \subset \mathbb{R}^{d+1}$ be a polytope with non-empty interior, $P = \pi(\tilde{P})$. A face of $\tilde{P}$ is *lower* if it is the set of maximisers of some linear functional $\nu = (\nu', \nu_{d+1})$ with $\nu_{d+1} < 0$ on $\tilde{P}$; a lower facet is a lower face of dimension d.

Lemma B.1 (lifting)**.** *(i) On every lower face G the projection π is an affine isomorphism onto $\pi(G)$; (ii) the sets $\pi(F)$, F a lower facet, cover P; (iii) the interiors of $\pi(F)$ and $\pi(F')$ are disjoint for distinct lower facets $F \neq F'$.*

Proof. Put $h(x) = \min\{t : (x,t) \in \tilde{P}\}$ for $x \in P$.

(i) Let G be the set of maximisers of ν with $\nu_{d+1} < 0$ and $(x,t) \in G$. If $(x,t') \in \tilde{P}$ with $t' < t$, then $\nu \cdot (x,t') - \nu \cdot (x,t) = \nu_{d+1}(t' - t) > 0$, contradicting maximality; hence $t = h(x)$. So $G \subseteq \{(x, h(x))\}$ and $\pi|_G$ is injective, and an injective affine map on a polytope is an affine isomorphism onto its image; in particular $\dim \pi(G) = \dim G \leq d$.

(ii) The set $\pi(\operatorname{int} \tilde{P})$ is open (since π is an open map), convex, and dense in P; for a convex set C with $\overline{C} = P$ one has $\operatorname{int} P = \operatorname{int} C$, so $\pi(\operatorname{int} \tilde{P}) = \operatorname{int} P$. Let $x \in \operatorname{int} P$ and $y = (x, h(x))$. The fibre $\tilde{P} \cap (\{x\} \times \mathbb{R})$ contains a point (x, t_0) of $\operatorname{int} \tilde{P}$, and $y \in \partial \tilde{P}$ by minimality of $h(x)$. Let ν be a supporting functional at y, $G_\nu = \{z \in \tilde{P} : \nu \cdot z = \nu \cdot y\}$. Since $(x,t) \in \tilde{P}$ for $t \geq h(x)$ near t_0 we have $\nu_{d+1} \leq 0$; and $\nu_{d+1} = 0$ would give $\nu \cdot (x, t_0) = \nu \cdot y$, i.e. an interior point of $\tilde{P}$ on a supporting hyperplane, which is impossible. Hence $\nu_{d+1} < 0$, so y lies in a lower face G_ν and $x \in \pi(G_\nu)$. Thus $\operatorname{int} P \subseteq \bigcup \pi(G)$ over lower faces G. The lower faces of dimension $< d$ project (by (i)) to finitely many polytopes of dimension $< d$, a closed set with empty interior; a lower face of dimension d is a lower facet. So the union of $\pi(F)$ over lower facets is a closed set that contains a dense subset of $\operatorname{int} P$, hence contains $\operatorname{int} P$ and, being closed, $P = \overline{\operatorname{int} P}$.

(iii) Let $F \neq F'$ be lower facets and x an interior point of both $\pi(F)$ and $\pi(F')$. By (i) the points of F, F' above x are both equal to $y = (x, h(x))$, so $y \in F \cap F'$; as $\pi|_F$, $\pi|_{F'}$ are affine isomorphisms onto d-dimensional sets and x is interior to the images, y lies in the relative interior of both F and F'. But a point of a polytope lies in the relative interior of exactly one face, so $F = F'$. □

Remark B.2. For the zonotope $\tilde{Z} = \sum [0,1]\tilde{v}_i$ of Lemma 7.2, full-dimensionality holds because, for $n > d$ and generic heights, some lifted generator is not in the span of the d others of a basis; the lemma then gives that the lower facets $\sum_{i \in A_S} \tilde{v}_i + Z(\tilde{v}_S)$ project onto tiles T_S that cover $Z(V)$ with disjoint interiors, which is the claim of Lemma 7.2.